MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SWITCH, CROSSPOINT, KEY CODE LZN

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 SCOPE. This document establishes the requirements and test procedures for the acquisition of an electronic crosspoint switch module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 2.620 failures per million operating hours for class I or 24.67 failures per million operating hours for class II when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirement specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be LZN.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/320-1 for class I and M28787/320-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I(see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

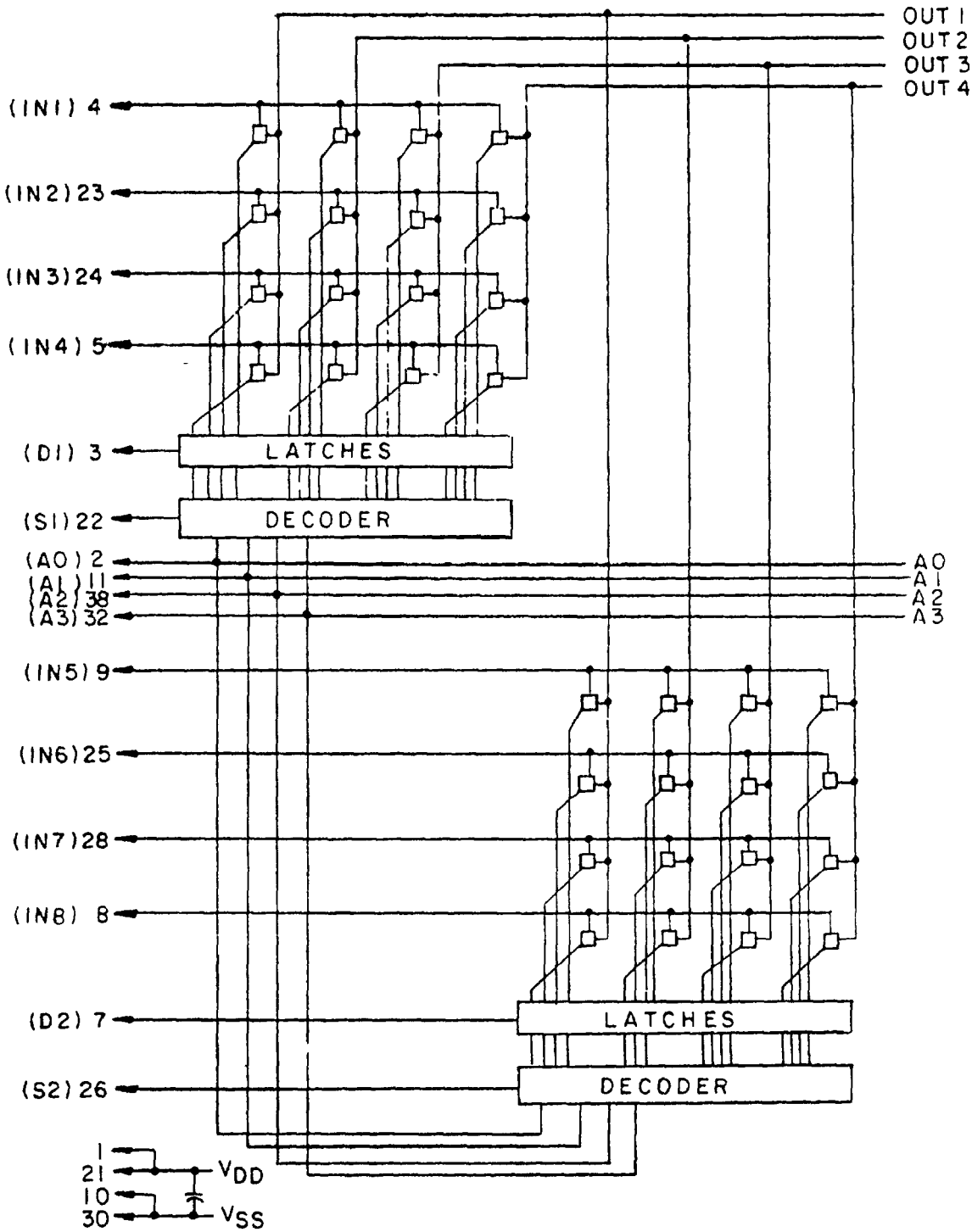
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

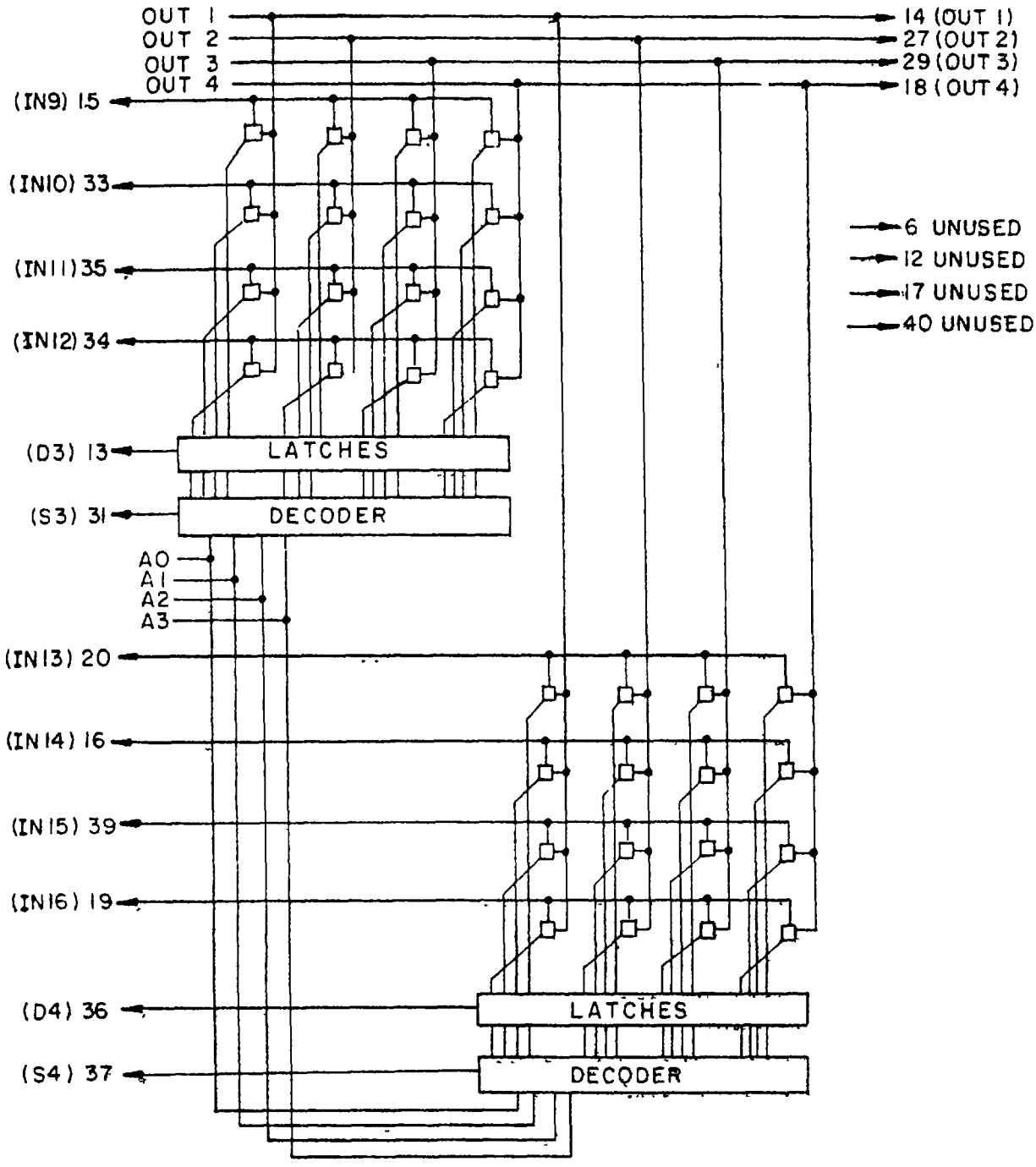
4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirement of MIL-M-28787.

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* FIGURE 1. LZN module function diagram, crosspoint switch.

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* FIGURE 1. LZN module function diagram, crosspoint switch - Continued.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A4	Stress	III	Q	16.0	18.2	16.0	18.2	15.8	18.2	16.0	18.2	15.5	18.2	V
B1-B5	Isolation													
	Initial	IV	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	IV	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
C1-C8	Negative input clamp diode voltage	V	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C9-C12	Negative input clamp diode voltage	V	X	-0.3	-2.0	-0.3	-2.0	-0.3	-2.0	-0.3	-2.0	-0.3	-2.0	V
D1-D8	Positive input clamp diode voltage	VI	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D9-D12	Positive input clamp diode voltage	VI	X	0.3	1.0	0.3	1.0	0.3	1.0	0.3	1.0	0.3	1.0	V
E1, E2	Continuity	VII	X	0	1	0	1	0	1	0	1	0	1	Ω
F1, F2	Power supply current	VIII	S	-0.01	22.00	-0.01	22.00	-0.01	660.00	-0.01	22.00	-0.01	660.00	µA
F3,F4	Power supply current	VIII	S	-0.01	88.00	-0.01	88.00	-0.01	2640.00	-0.01	88.00	-0.01	2640.00	µA
G1-G64	Input functional	IX	X	8.7	11.1	8.7	11.1	8.5	11.1	8.7	11.1	8.4	11.6	mA
H1-H4	Output functional	X	X	8.7	11.1	8.7	11.1	8.5	11.1	8.7	11.1	8.4	11.6	mA
J1-J8	Leakage current, input and output	XI	X	0.1	200.0	---	---	1	2,000	---	---	10	4,000	nA
K1-K64	R_{ON} (V_{DD}-V_{SS} = 4.5 V) check	XII	X	10	700	10	700	10	800	10	700	10	850	Ω
L1-L32	R_{ON} (variable input voltage) check	XIII	X	10	700	10	700	10	800	10	700	10	850	Ω
M1-M4	AC feedthrough check	XIV	X	-∞	-34	-∞	-34	-∞	-34	-∞	-34	-∞	-34	dB

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N1-N4	Large signal response	XV	X	4.4	4.7	4.4	4.7	4.4	4.7	4.4	4.7	4.4	4.7	V rms
P1-P4	Output crosstalk check	XVI	Q	-∞	-24.3	---	---	-∞	-24.3	---	---	-∞	-24.3	dB
Q1-Q4	Input crosstalk check	XVII	Q	-∞	-24.3	---	---	-∞	-24.3	---	---	-∞	-24.3	dB
R1-R12	Control crosstalk (address, data-in, and strobe)	XVIII	Q	-450	450	---	---	-450	450	---	---	-450	450	mV
S1-S12	Input current, high level	XIX	X	0	200	---	---	0	500	---	---	0	1,000	nA
T1-T12	Input current, low level	XX	X	0	200	---	---	0	500	---	---	0	1,000	nA
U	Power supply filter capacitance	XXI	X	0.075	0.125	0.075	0.125	0.075	0.125	0.070	0.130	0.070	0.130	µF
V1-V12	Input capacitance, digital	XXII	Q	2	25	2	25	2	25	2	25	2	25	pF
V13-V28	Input capacitance, switch	XXII	Q	2	60	2	60	2	60	2	60	2	60	pF
V29-V32	Output capacitance, switch	XXII	Q	10	300	10	300	10	300	10	300	10	300	pF
W1-W4	Propagation delay time, address to output (t_{PHZ}), V_{DD} = 5 V	XXIII	S	100	1000	100	1000	120	1200	100	1000	130	1300	ns
X1-X4	Propagation delay time, address to output (t_{PZH}), V_{DD} = 5 V	XXIV	S	70	700	67	670	77	770	53	530	88	880	ns
Y1-Y64	Propagation delay time, strobe to output (t_{PHZ}), V_{DD} = 5 V	XXV	S	70	700	67	670	77	770	53	530	88	880	ns
Z1-Z64	Propagation delay time, strobe to output (t_{PZH}), V_{DD} = 5 V	XXVI	S	60	600	57	570	66	660	45	450	75	750	ns
AA1-AA4	Propagation delay time, data-in to output (t_{PHZ}), V_{DD} = 5 V	XXVII	S	60	600	57	570	66	660	45	450	75	750	ns
---														

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min > Max		Min	Max	Min	Max	Min	Max	
BB1-BB4	Propagation delay time, data-in to output (t_{PZH}), V_{DD} = 5 V	XXVIII	S	60	600	57	570	66	660	45	450	75	750	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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* 4.4.2 Order of testing. Test conditions A through BB shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through BB in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through H, K through N, and U through BB shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. Test conditions J and P through T shall be performed at 25°C, 60°C and 85°C using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

* 4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column except test conditions C and D shall be performed using the initial limits at 25°C ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E through BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests E through BB shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation, V_{DD} = 5.5 V	13.75	14.3	mW
V_{DD} = 16.5 V	190	198	mW
Input voltage			
V_{DD} = 5 V			
Low level	1.3 max	1.3 max	V
High level	3.9 min	3.9 min	V
V_{DD} = 15 V			
Low level	3.4 max	3.4 max	V
High level	12.4 min	12.4 min	V
Setup time (data-in to strobe)			
V_{DD} = 5 V	200	200	ns
Hold time (data-in to strobe)			
V_{DD} = 5 V	415	415	ns
Minimum strobe pulse width			
V_{DD} = 5 V	620	620	ns

6.5 Module function. This module contains a 16 input lines to 4 output lines analog switching matrix, consisting of 64 analog switches. The switches are controlled by 12 digital inputs.

6.5.1 Input to output relationship. The digital input versus analog switch selection is shown in table II.

6.6 Module handling considerations. To avoid possible damage to the devices during handling, testing or actual operation, the following procedures should be followed.

6.6.1 Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid build-up of static charge.

6.6.2 Soldering iron tips, metal parts of fixtures and tools, test equipment and handling facilities should be grounded.

6.6.3 Signals should not be applied to the inputs while the device power supply is off.

6.6.4 Input voltages greater than the supply (V_{DD} - V_{SS}) should not be applied as this will forward bias the protection input diodes.

6.6.5 All unused input leads must be connected to either V_{SS} (ground) or V_{DD} (device supply), whichever is appropriate for the logic circuit involved.

6.6.6 CAUTION. Failure to comply with the following procedures during testing will result in irreversible module damage.

6.6.6.1 All test equipment must have the chassis connected to earth ground.

6.6.6.2 V_{DD} must be applied before applying input signals or conditioning voltages.

6.6.6.3 All input signals and conditioning voltages must be removed before V_{DD} is disconnected.

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6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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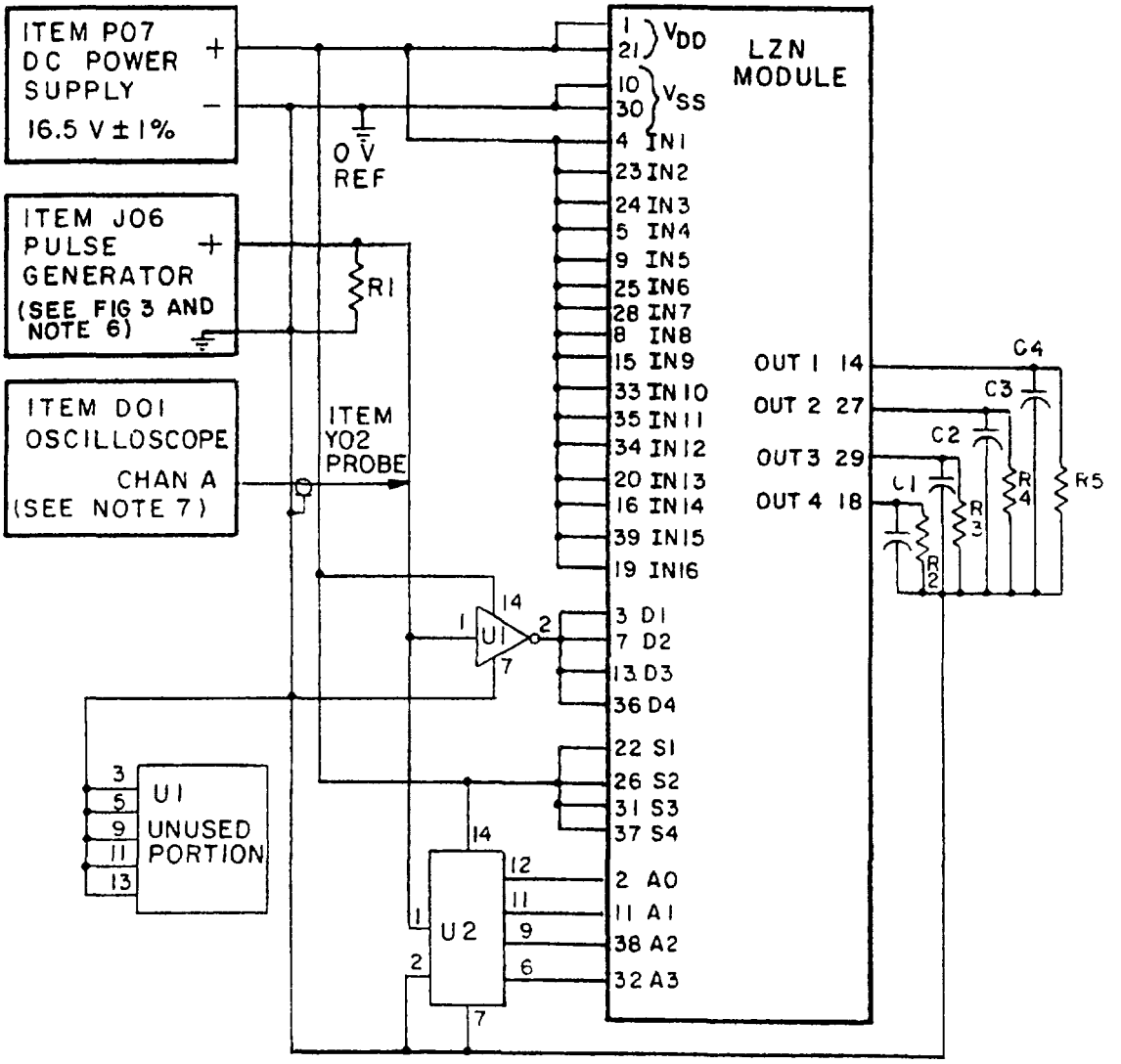
TABLE II. Analog switch selection. 1/ 2/

Address (module pin number)				Data-in and strobe (module pin numbers)								Output selected
A0 (2)	A1 (11)	A2 (38)	A3 (32)	D1 (3) S1 (22)		D2 (7) S2 (26)		D3 (13) S3 (31)		D4 (36) S4 (37)		(module pin number)
L	L	L	L	IN1	(4)	IN5	(9)	IN9	(15)	IN13	(20)	Out 1 (14)
H	L	L	L	IN2	(23)	IN6	(25)	IN10	(33)	IN14	(16)	"
L	H	L	L	IN3	(24)	IN7	(28)	IN11	(35)	IN15	(39)	"
H	H	L	L	IN4	(5)	IN8	(8)	IN12	(34)	IN16	(19)	"
L	L	H	L	IN1	(4)	IN5	(9)	IN9	(15)	IN13	(20)	Out 2 (27)
H	L	H	L	IN2	(23)	IN6	25)	IN10	(33)	IN14	(16)	"
L	H	H	L	IN3	(24)	IN7	(28)	IN11	(35)	IN15	(39)	"
H	H	H	L	IN4	(5)	IN8	(8)	IN12	(34)	IN16	(19)	"
L	L	L	H	IN1	(4)	IN5	(9)	IN9	(15)	IN13	(20)	Out 3 (29)
H	L	L	H	IN2	(23)	IN6	(25)	IN10	(33)	IN14	(16)	"
L	H	L	H	IN3	(24)	IN7	(28)	IN11	(35)	IN15	(39)	"
H	H	L	H	IN4	(5)	IN8	(8)	IN12	(34)	IN16	(19)	"
L	L	H	H	IN1	(4)	IN5	(9)	IN9	(15)	IN13	(20)	Out 4 (18)
H	L	H	H	IN2	(23)	IN6	(25)	IN10	(33)	IN14	(16)	"
L	H	H	H	IN3	(24)	IN7	(28)	IN11	(35)	IN15	(39)	"
H	H	H	H	IN4	(5)	IN8	(8)	IN12	(34)	IN16	(19)	"

1/ L = logic low level D = data-in
H = logic high level S = strobe
A = address

2/ The required switch is selected by applying the address and placing a high at the applicable strobe input. The switch is turned on if the applicable data-in is high or turned off if the applicable data-in is low. The switch is latched at the new position.

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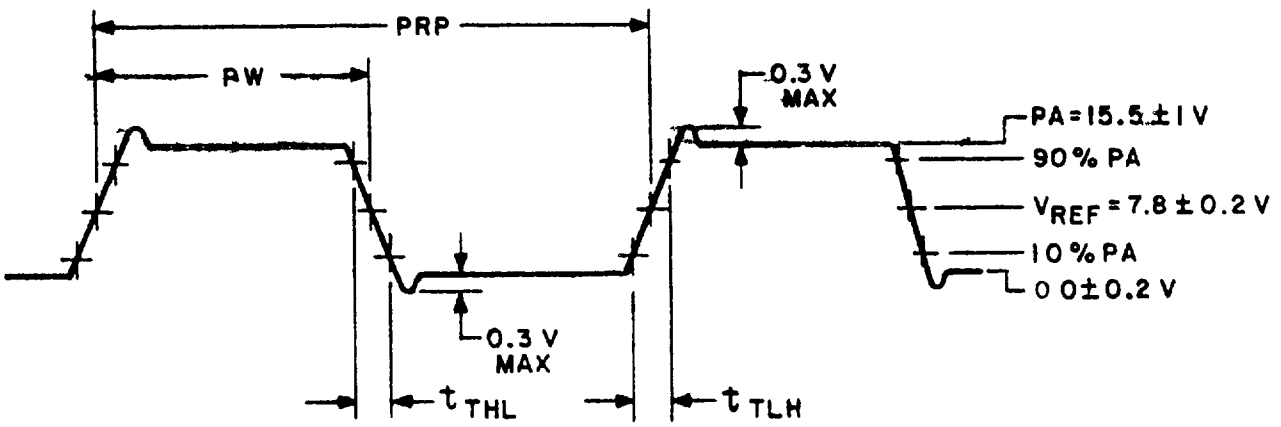
CAUTION:

Observe the procedures of 6.6.6.

NOTES:
1. R1 = 51.1Ω ±5%, 5 W.
2. R2-R5 = 1.82 kΩ ±1%, 1/4 W.
3. C1-C4 = 47 pF ±10%.
4. U1 = CD4069UB, or equivalent.
5. U2 = CD4024B, or equivalent.
6. Adjust item J06 pulse prior to applying pulse to the module under test.
7. Item D01 is used for setup purposes only and can be removed during life testing.

FIGURE 2. Operating temperature, component temperature, and life test circuit.

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NOTE: Input pulse parameters:
PRP = 1.0 ±0.2 µs.
PW = 0.5 ±0.1 µs.
$t_{TLH} = t_{THL}$ = 50 ns maximum.

FIGURE 3. Operating temperature, component temperature, and life test circuit waveform.

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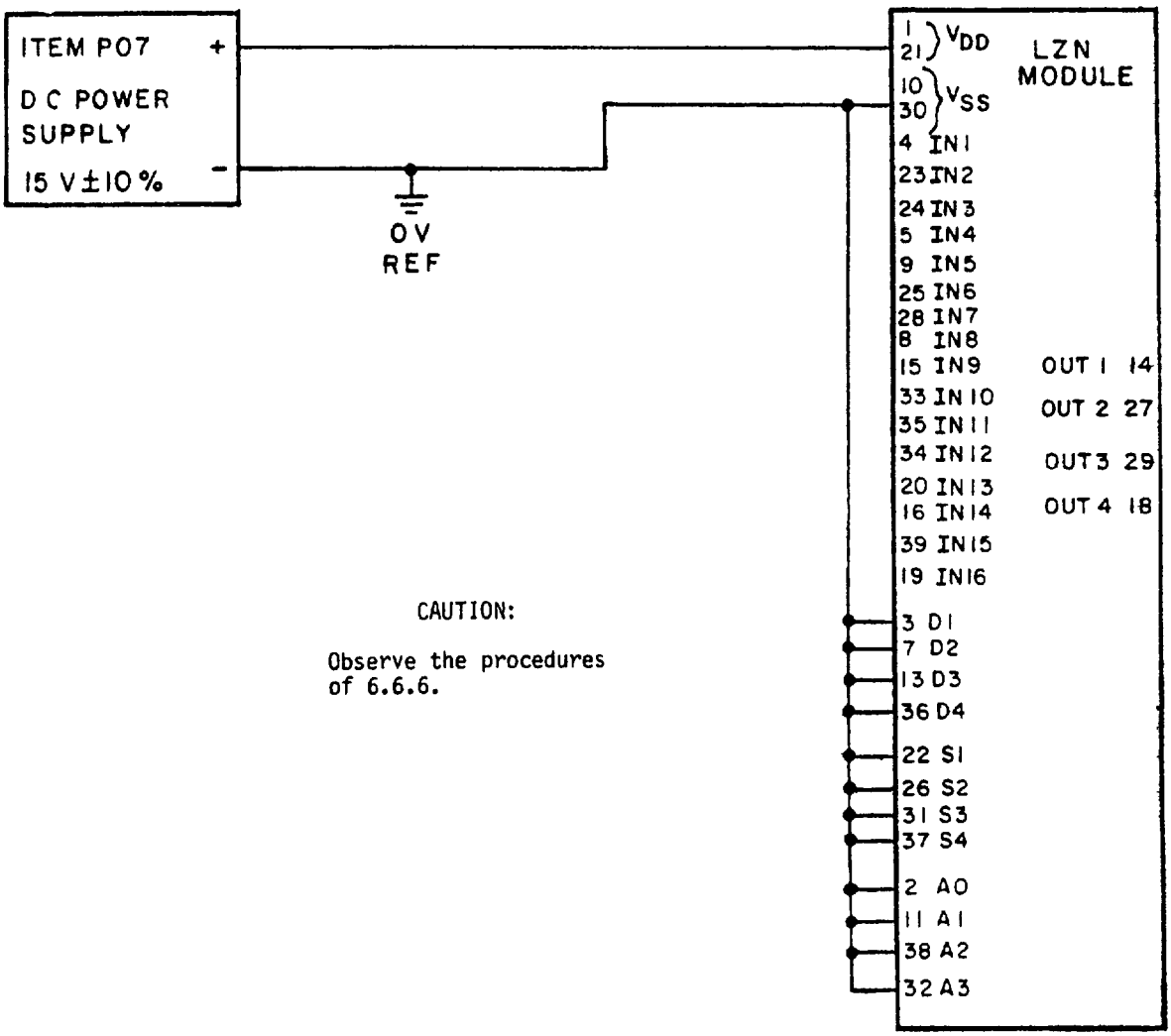


FIGURE 4. Vibration test circuit.

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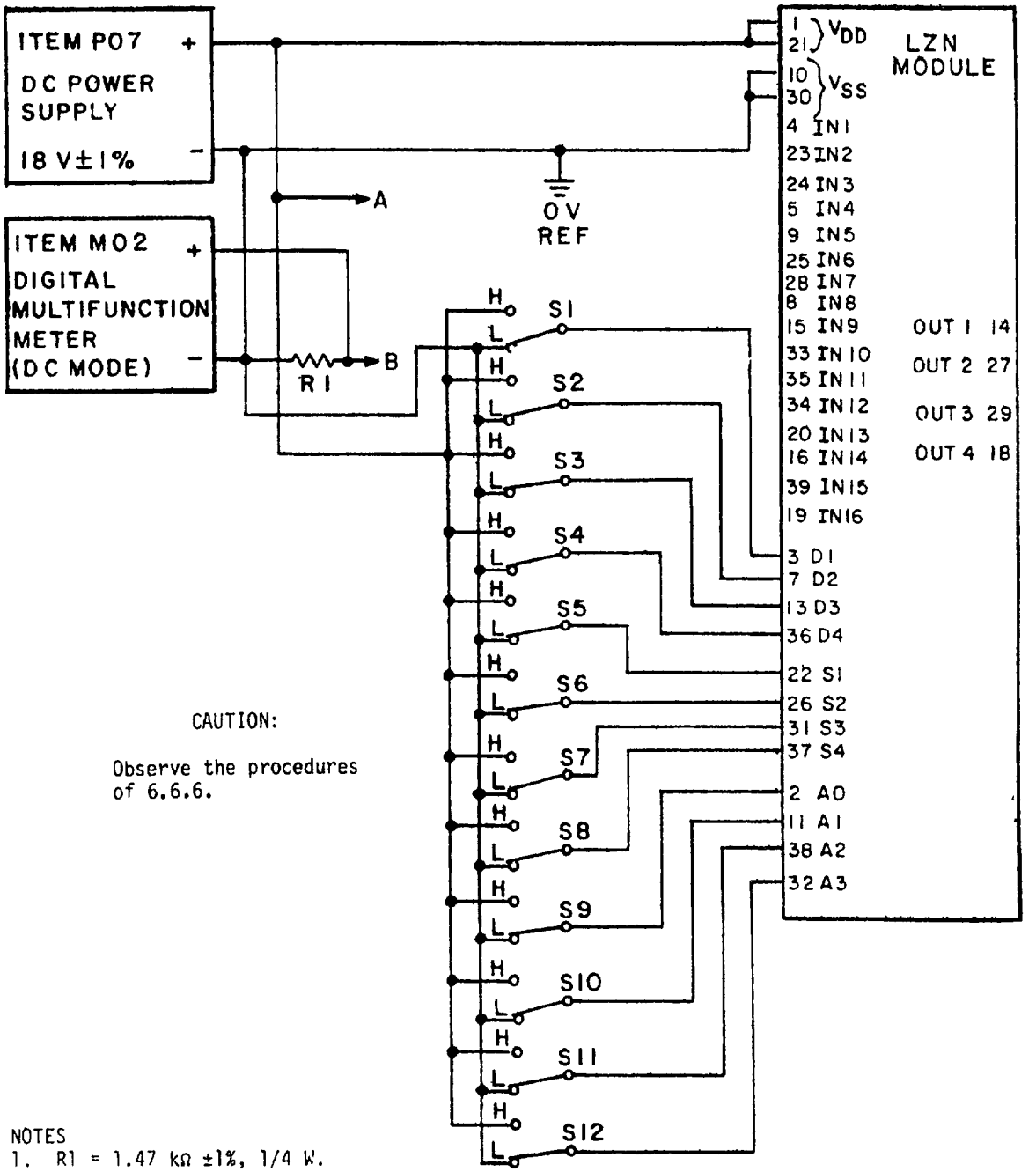
TABLE III. Stress test.

Test condition	Parameter	Channel ON switch	Address switch				Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			S9	S10	S11	S12	A to pin	B to pin	Min	Max		
A1	Stress	S4	H	H	L	L	19	14	16.0	18.2	V	5
A2	"	S3	H	H	H	L	34	27	"	"	"	"
A3	"	S2	H	H	L	H	8	29	"	"	"	"
A4	"	S1	H	H	H	H	5	18	"	"	"	"

TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Isolation	C	Frame	10	∞	MΩ	7
B2	"	C	6,12,17,40	"	"	"	"
B3	"	6	12,17,40	"	"	"	"
B4	"	12	17,40	"	"	"	"
B5	"	17	40	"	"	"	"

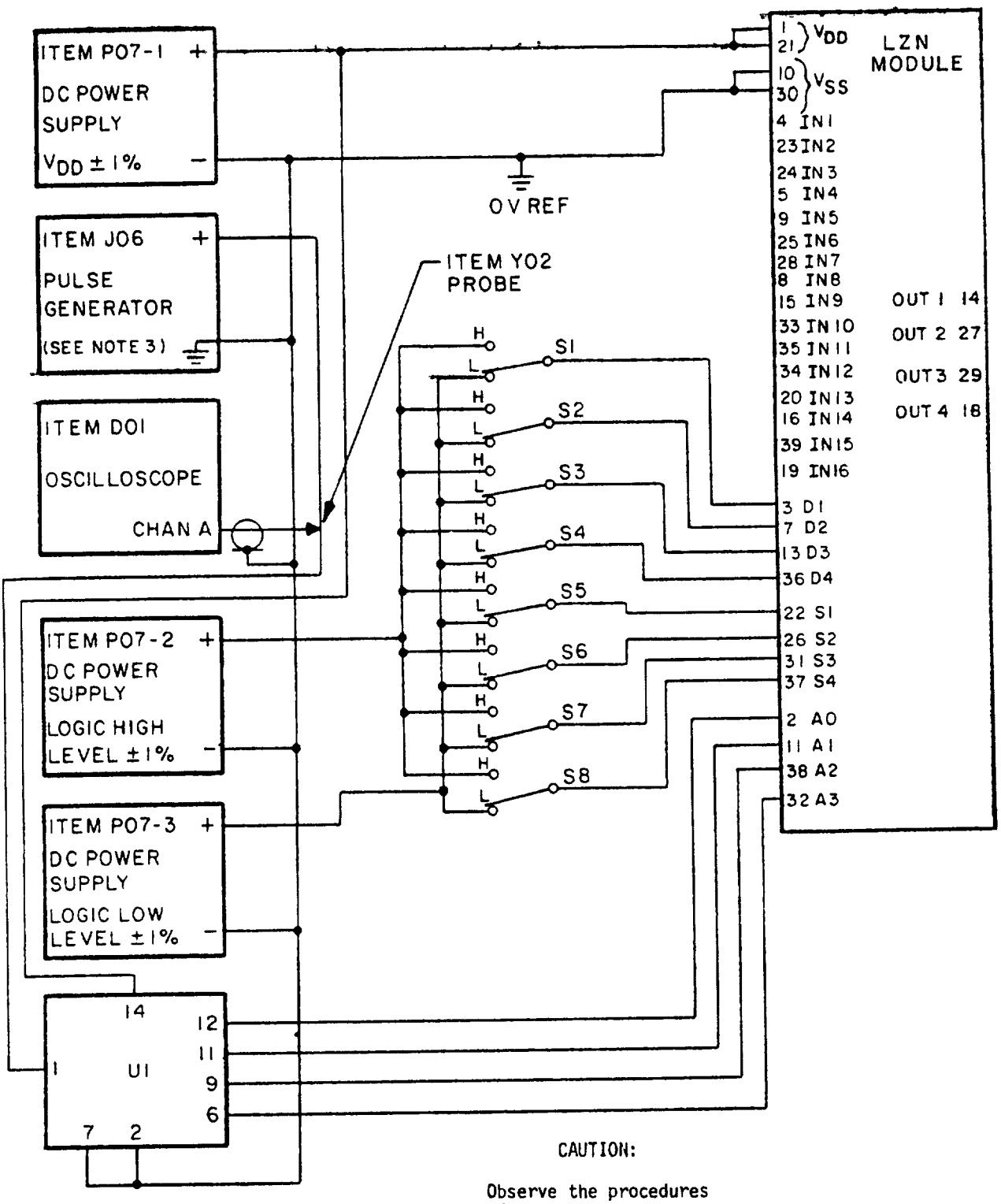
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NOTES
1. R1 = 1.47 kΩ ±1%, 1/4 W.
2. S1-S12 = SPDT switch.
3. When power is initially applied the analog switches may be either OFF or ON. Before performing this test set all switches OFF by performing the procedure of figure 6 (set V_{DD} = 18 V, logic high level = 18 V, and logic low level = 0 V).
4. Perform test condition A1 through A4 as follows:
   a. Do not de-energize item P07 during the remainder of the tests.
   b. Keep S1-S4 in position L (except as noted) and switch S5-S8 to H.
   c. Switch S9-S12 to the correct position per table III.
   d. Connect points A and B and switch the applicable channel ON switch from position L to position H (see table III).
   e. Wait one minute and read item M02 voltage.
   f. Switch the applicable channel ON switch back to position L.
   g. Repeat steps c through f for each test condition.

FIGURE 5 Stress test circuit for table III.

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NOTES:
1. S1-S8 = SPDT switch.
2. U1 = CD4024B, or equivalent.

FIGURE 6 Test circuit for turning all analog switches to the OFF state (single power supply).

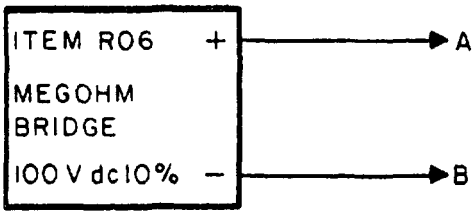
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NOTES: (continued)

3. Input pulse parameters:
   PRP = 4 µs.
   PW = 2 µs.
   $t_{TLH} = t_{THL}$ = 50 ±40 ns.
   Pulse high level = Same as V_{DD} +0, -10% (module will be damaged if pulse high level is greater than V_{DD}).
   Pulse low level = 0.0 +0.4, -0.0 V (module will be damaged if pulse low level is less than V_{SS}).
4. Procedure:
   a. Set switches S1-S4 to position L.
   b. Set switches S5-S8 to position H.
   c. Turn on item P07-1 and then connect item J06 and turn on items P07-2 and P07-3. This will turn all analog switches OFF after U1 counts through all possible states (this will take approximately 64 µs).
   d. Set switches S5-S8 to position L (this will disconnect the address pins 2, 11, 32, and 38 from the analog switches while disconnecting U1).
   e. Return to the applicable test condition while keeping item P07 energized.

FIGURE 6. Test circuit for turning all analog switches to the OFF state (single power supply) - Continued

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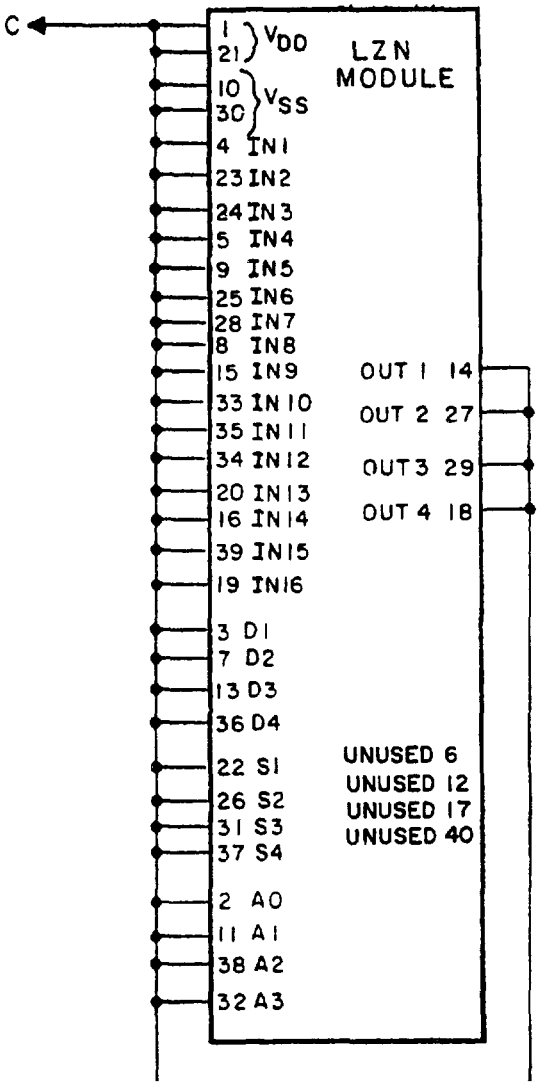
CAUTION:

Observe the procedures of 6.6.6.

WARNING

When using a megohm bridge, high voltage may be present at any of the binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting the test leads.

NOTE When performing test condition B1 penetrate the anodized finish of the module frame with



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TABLE V. Negative input clamp diode voltage test.

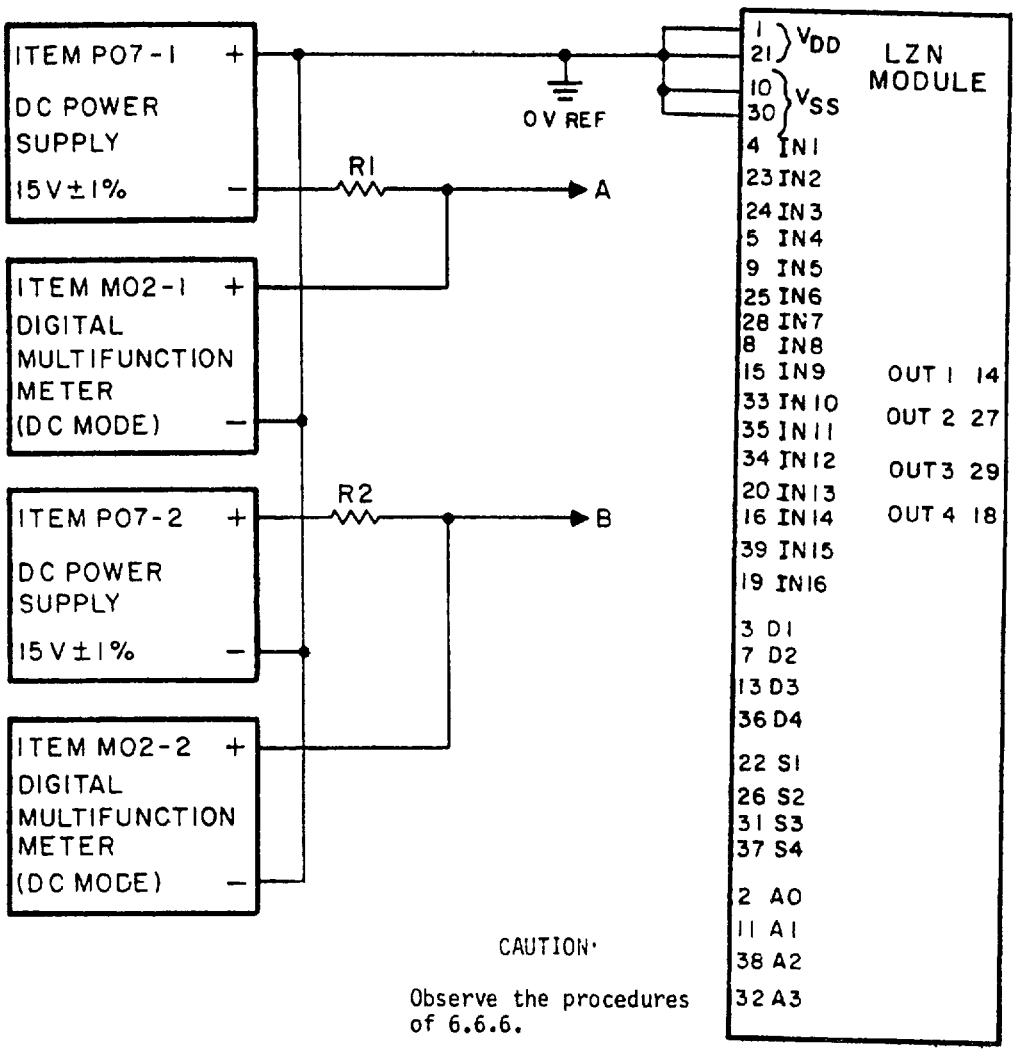
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	D1	Negative input clamp diode voltage	3	Open	-0.4	-6.0	V	8
C2	D2	"	7	"	"	"	"	"
C3	D3	"	13	"	"	"	"	"
C4	D4	"	36	"	"	"	"	"
C5	S1	"	22	"	"	"	"	"
C6	S2	"	26	"	"	"	"	"
C7	S3	"	31	"	"	"	"	"
C8	S4	"	37	"	"	"	"	"
C9	A0	"	2	"	-0.3	-2.0	"	"
C10	A1	"	11	"	"	"	"	"
C11	A2	"	38	"	"	"	"	"
C12	A3	"	32	"	"	"	"	"

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TABLE VI. Positive input clamp diode voltage test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	D1 Positive input clamp diode voltage	Open	3	0.4	1.5	V	8
D2	D2 "	"	7	"	"	"	"
D3	D3 "	"	13	"	"	"	"
D4	D4 "	"	36	"	"	"	"
D5	S1 "	"	22	"	"	"	"
D6	S2 "	"	26	"	"	"	"
D7	S3 "	"	31	"	"	"	"
D8	S4 "	"	37	"	"	"	"
D9	A0 "	"	2	0.3	1.0	"	"
D10	A1 "	"	11	"	"	"	"
D11	A2 "	"	38	"	"	"	"
D12	A3 "	"	32	"	"	"	"

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NOTES
1. R1 = 9.09 kΩ ±1%, 1/8 W.
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3. Since items P07-1, P07-2 and M02-1, M02-2 are not used simultaneously, only one item P07 and M02 is required.

* FIGURE 8. Negative and positive input clamp diode voltage test circuit for tables V and VI.

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TABLE VII. Continuity test.

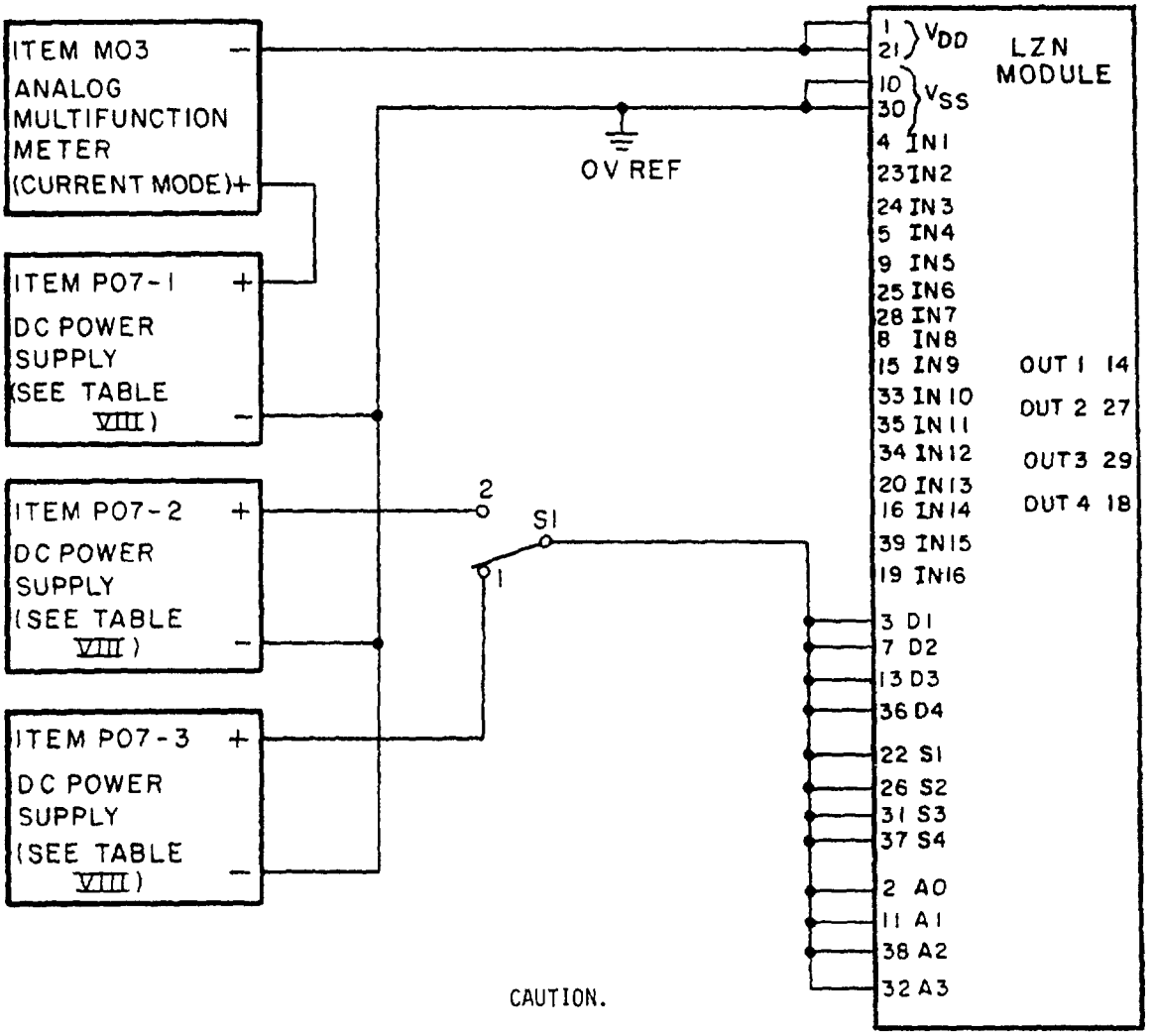
Test condition	Parameter	Connect Item M02 1/		Initial limits 25°C ±5°C ambient		Unit	Test figure
		+ terminal to pin	- terminal to pin	Min	Max		
E1	Continuity	1	21	0	1	Ω	---
E2	Continuity	10	30	0	1	Ω	

1/ Item M02 shall be in the ohms mode.

* TABLE VIII. Power supply current test.

Test condition	Parameter	Item P07-2 voltage ±1%	Item P07-3 voltage ±1%	Item P07-1 voltage ±1%	S1 position	Initial limits 25°C ±5°C ambient		Unit	Test figure
						Min	Max		
F1	Power supply current	---	0.10	5.5	1	-0.01	22.00	µA	9
F2	"	5.40	---	5.5	2	-0.01	22.00	"	"
F3	"	---	0.10	16.5	1	-0.01	88.00	"	"
F4	"	16.4	---	16.5	2	-0.01	88.00	"	"

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CAUTION.

Observe the procedures of 6.6.6.

WARNING:

Item M03 shall be in the current mode when item P07-1 is turned on or else irreversible module damage will result.

NOTE: S1 = SPDT switch.

FIGURE 9. Power supply current test circuit for table VIII.

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TABLE IX. Input functional test.

Test condition	Parameter	Channel ON switch	Address switch				Initial limits 25°C ±5°C ambient		Unit	Test figure
			S9	S10	S11	S12	Min	Max		
G1	Input functional	S1	L	L	L	L	8.7	11.1	mA	10
G2	"	S2	"	"	"	"	"	"	"	"
G3	"	S3	"	"	"	"	"	"	"	"
G4	"	S4	"	"	"	"	"	"	"	"
G5	"	S1	H	"	"	"	"	"	"	"
G6	"	S2	"	"	"	"	"	"	"	"
G7	"	S3	"	"	"	"	"	"	"	"
G8	"	S4	"	"	"	"	"	"	"	"
G9	"	S1	L	H	"	"	"	"	"	"
G10	"	S2	"	"	"	"	"	"	"	"
G11	"	S3	"	"	"	"	"	"	"	"
G12	"	S4	"	"	"	"	"	"	"	"
G13	"	S1	H	"	"	"	"	"	"	"
G14	"	S2	"	"	"	"	"	"	"	"
G15	"	S3	"	"	"	"	"	"	"	"
G16	"	S4	"	"	"	"	"	"	"	"
G17	"	S1	L	L	H	"	"	"	"	"
G18	"	S2	"	"	"	"	"	"	"	"
G19	"	S3	"	"	"	"	"	"	"	"
G20	"	S4	"	"	"	"	"	"	"	"
G21	"	S1	H	"	"	"	"	"	"	"
G22	"	S2	"	"	"	"	"	"	"	"

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TABLE IX. Input functional test - Continued.

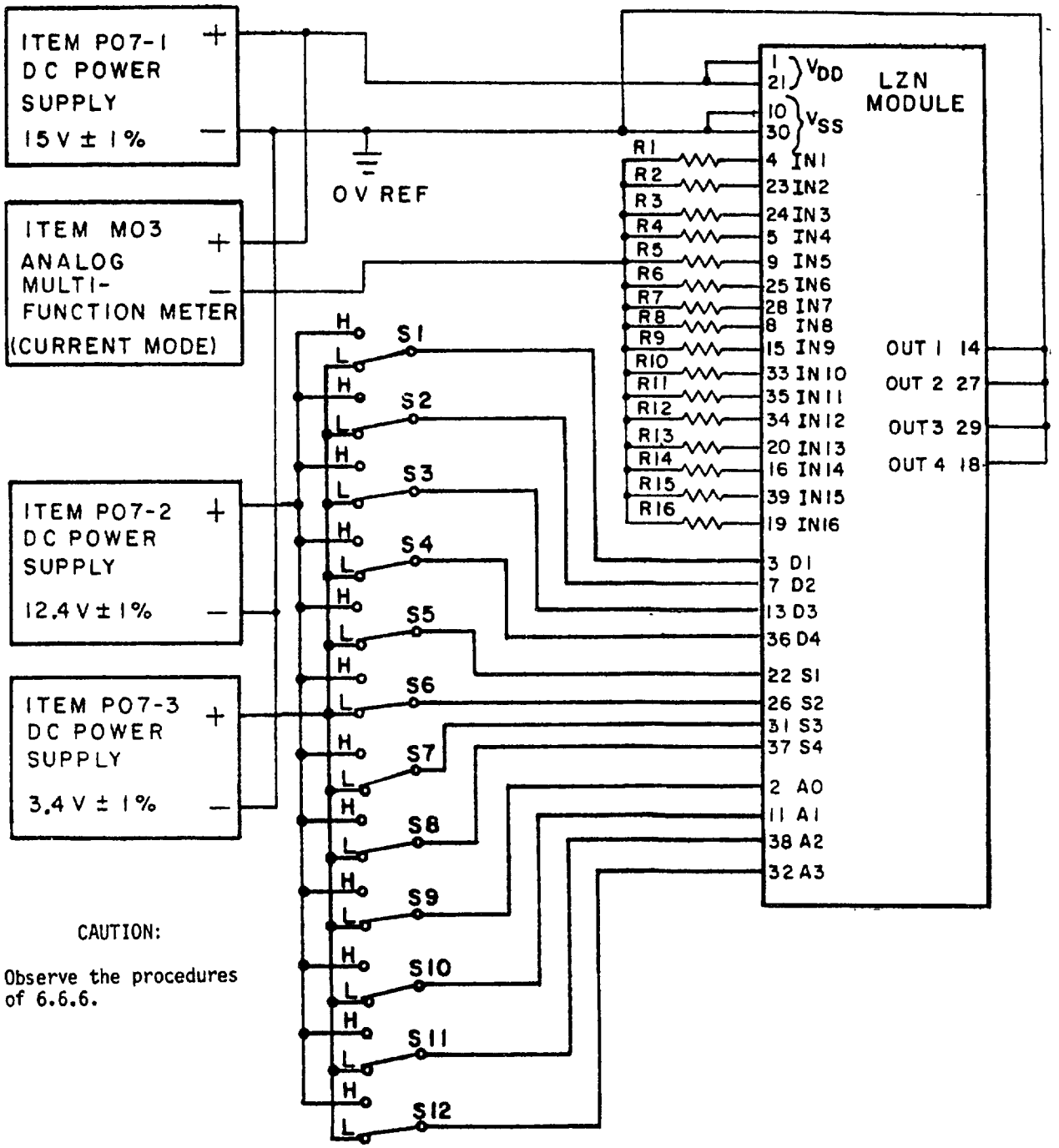
Test condition	Parameter	Channel ON switch	Address switch				Initial limits 25°C ±5°C ambient		Unit	Test figure
			S9	S10	S11	S12	Min	Max		
G23	Input functional	S3	H	L	H	L	8.7	11.1	mA	10
G24	"	S4	H	L	"	"	"	"	"	"
G25	"	S1	L	H	"	"	"	"	"	"
G26	"	S2	"	"	"	"	"	"	"	"
G27	"	S3	"	"	"	"	"	"	"	"
G28	"	S4	"	"	"	"	"	"	"	"
G29	"	S1	H	"	"	"	"	"	"	"
G30	"	S2	"	"	"	"	"	"	"	"
G31	"	S3	"	"	"	"	"	"	"	"
G32	"	S4	"	"	"	"	"	"	"	"
G33	"	S1	L	L	L	H	"	"	"	"
G34	"	S2	"	"	"	"	"	"	"	"
G35	"	S3	"	"	"	"	"	"	"	"
G36	"	S4	"	"	"	"	"	"	"	"
G37	"	S1	H	"	"	"	"	"	"	"
G38	"	S2	"	"	"	"	"	"	"	"
G39	"	S3	"	"	"	"	"	"	"	"
G40	"	S4	"	"	"	"	"	"	"	"
G41	"	S1	L	H	"	"	"	"	"	"
G42	"	S2	"	"	"	"	"	"	"	"
G43	"	S3	"	"	"	"	"	"	"	"
G44	"	S4	"	"	"	"	"	"	"	"

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TABLE IX. Input functional test - Continued.

Test condition	Parameter	Channel ON switch	Address switch				Initial limits 25°C ±5°C ambient		Unit	Test figure
			S9	S10	S11	S12	Min	Max		
G45	Input functional	S1	H	L	L	H	8.7	11.1	mA	10
G46	"	S2	"	"	"	"	"	"	"	"
G47	"	S3	"	"	"	"	"	"	"	"
G48	"	S4	"	"	"	"	"	"	"	"
G49	"	S1	L	L	H	"	"	"	"	"
G50	"	S2	"	"	"	"	"	"	"	"
G51	"	S3	"	"	"	"	"	"	"	"
G52	"	S4	"	"	"	"	"	"	"	"
G53	"	S1	H	"	"	"	"	"	"	"
G54	"	S2	"	"	"	"	"	"	"	"
G55	"	S3	"	"	"	"	"	"	"	"
G56	"	S4	"	"	"	"	"	"	"	"
G57	"	S1	L	H	"	"	"	"	"	"
G58	"	S2	"	"	"	"	"	"	"	"
G59	"	S3	"	"	"	"	"	"	"	"
G60	"	S4	"	"	"	"	"	"	"	"
G61	"	S1	H	"	"	"	"	"	"	"
G62	"	S2	"	"	"	"	"	"	"	"
G63	"	S3	"	"	"	"	"	"	"	"
G64	"	S4	"	"	"	"	"	"	"	"

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* FIGURE 10. Input functional test circuit for table IX.

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NOTES:
1. R1-R16 = 1.47 kΩ ±1%, 1/4 W.
2. S1-S12 = SPDT switch.
3. When input power is initially applied, the analog switches may be either OFF or ON To set all switches OFF perform the procedure of figure 6 (set V_{DD} = 15 V, logic high level = 12.4 V, and logic low level = 3.4 V). This procedure shall be performed prior to setup per this figure.
4. The functional test shall be performed as follows:

   a. Do not de-energize items P07-1 through P07-3 while changing from the procedure of note 3 or during the remainder of the tests (if these are de-energized, the procedure of note 3 will have to be repeated).
   b. Keep S1-S4 in position L (except as noted) and switch S5-S8 to H.
   c. Switch S9-S12 to the correct position per table IX.
   d. Switch the channel ON switch specified in table IX from L to H. (This will turn the channel on and item M03 will change from less than 1 µA (for reference only) to a within limits reading).
   e. Read item M03 (use 0.1 ampere range).
   f. Switch the channel ON switch from H to L (this will turn the channel off).
   g. Repeat steps c through f for each test condition of table IX.

* FIGURE 10. Input functional test circuit for table IX - Continued.

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TABLE X. Output functional test.

Test condition	Parameter	Channel ON switch	Address switch				Initial limits 25°C ±5°C ambient		Unit	Test figure
			S9	S10	S11	S12	Min	Max		
H1	Output functional	S1	L	L	L	L	8.7	11.1	mA	11
H2	"	S2	"	"	H	L	"	"	"	"
H3	"	S3	"	"	L	H	"	"	"	"
H4	"	S4	"	"	H	H	"	"	"	"

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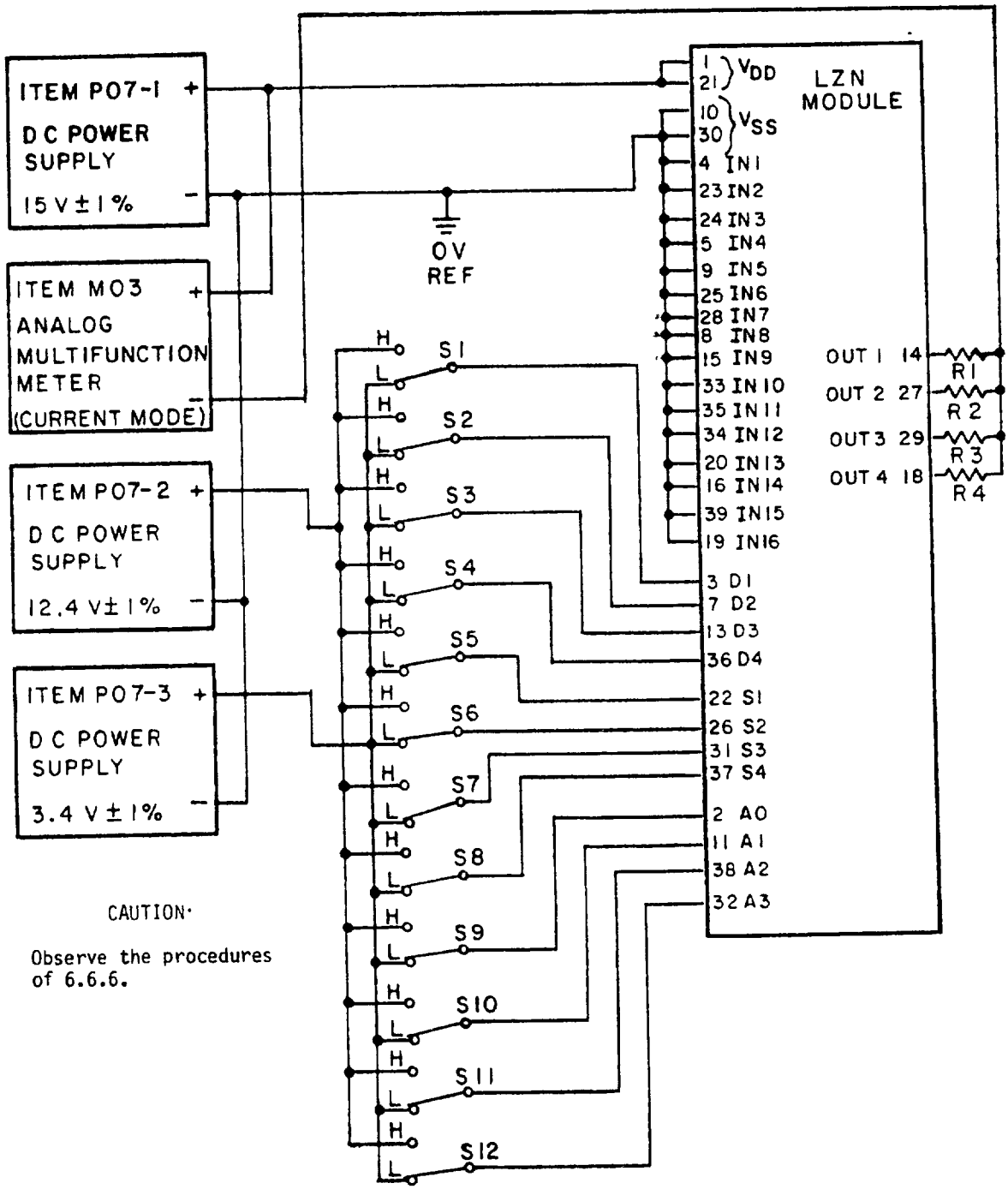


FIGURE 11. Output functional test circuit for table X.

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NOTES:
1. R1-R4 = 1.47 kΩ ±1%, 1/4 W.
2. S1-S12 = SPDT switch.
3. When input power is initially applied, the analog switches may be either OFF or ON. To set all switches OFF perform the procedure of figure 6 (set V_{DD} = 15 V, logic high level = 12.4 V, and logic low level = 3.4 V). This procedure shall be performed prior to setup per this figure.
4. The functional test shall be performed as follows:

   a. Do not de-energize items P07-1 through P07-3 while changing from the procedure of note 3 or during the remainder of the tests (if these are de-energized the procedure of note 3 will have to be repeated).
   b. Keep S1-S4 in position L (except as noted) and switch S5-S8 to H.
   c. Switch S9-S12 to the correct position per table X.
   d. Switch the channel ON switch specified in table X from L to H. This will turn the channel on and item M03 will change from less than 1 µA (for reference only) to a within limits reading.
   e. Read item M03.
   f. Switch the channel ON switch from H to L (this will turn the channel off).
   g. Repeat steps c through f for each test condition of table X.

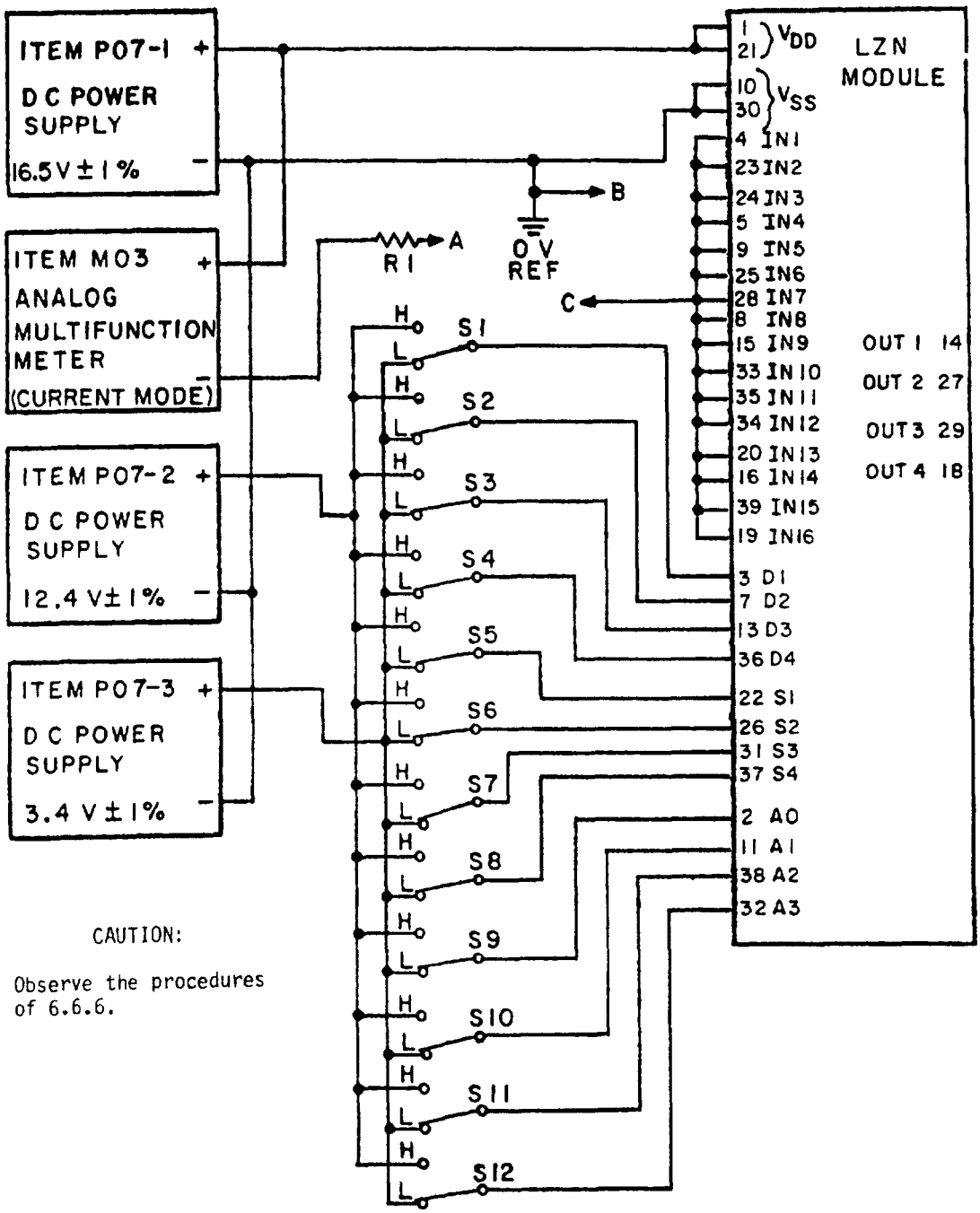
FIGURE 11. Output functional test circuit for table X - Continued.

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TABLE XI. Leakage current (input and output) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Leakage current, input	C	14	0.1	200.0	nA	12
J2	"	"	18	"	"	"	"
J3	"	"	27	"	"	"	"
J4	"	"	29	"	"	"	"
J5	Leakage current, output	14	C	"	"	"	"
J6	"	18	"	"	"	"	"
J7	"	27	"	"	"	"	"
J8	"	29	"	"	"	"	"

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NOTES:
1. R1 = 200 kΩ ±1%, 1/8 W.
2. S1-S12 = SPDT switch.
3. When power is initially applied, the analog switches may be either OFF or ON. To set all switches OFF, perform the procedure of figure 6 (set logic high level = 12.4 V, logic low level = 3.4 V, and V_{DD} = 16.5 V).
4. Before performing the tests, set switches S9-S12 in the L position (switches S1-S8 should already be in the L position). Without de-energizing item P07-1 make connections per table XI and read item M03 current for each test condition.

FIGURE 12. Leakage current (input and output) test circuit for table XI.

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TABLE XII. R_{ON} (V_{DD}-V_{SS} = 4.5 V) check test.

Test condition	Parameter	Channel ON switch	Address switch				Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			S9	S10	S11	S12	A to pin	B to pin	Min	Max		
K1	R_{ON} (V_{DD}-V_{SS} = 4.5 V) check 1/	S1	L	L	L	L	4	14	1	70	mV	13
K2	"	S2	"	"	"	"	9	"	"	"	"	"
K3	"	S3	"	"	"	"	15	"	"	"	"	"
K4	"	S4	"	"	"	"	20	"	"	"	"	"
K5	"	S1	H	"	"	"	23	"	"	"	"	"
K6	"	S2	"	"	"	"	25	"	"	"	"	"
K7	"	S3	"	"	"	"	33	"	"	"	"	"
K8	"	S4	"	"	"	"	16	"	"	"	"	"
K9	"	S1	L	H	"	"	24	"	"	"	"	"
K10	"	S2	"	"	"	"	28	"	"	"	"	"
K11	"	S3	"	"	"	"	35	"	"	"	"	"
K12	"	S4	"	"	"	"	39	"	"	"	"	"
K13	"	S1	H	"	"	"	5	"	"	"	"	"
K14	"	S2	"	"	"	"	8	"	"	"	"	"
K15	"	S3	"	"	"	"	34	"	"	"	"	"
K16	"	S4	"	"	"	"	19	"	"	"	"	"
K17	"	S1	L	L	H	"	4	27	"	"	"	"
K18	"	S2	"	"	"	"	9	"	"	"	"	"
K19	"	S3	"	"	"	"	15	"	"	"	"	"
K20	"	S4	"	"	"	"	20	"	"	"	"	"
K21	"	S1	H	"	"	"	23	"	"	"	"	"
K22	"	S2	"	"	"	"	25	"	"	"	"	"
K23	"	S3	"	"	"	"	33	"	"	"	"	"
K24	"	S4	"	"	"	"	16	"	"	"	"	"

See footnote at end of table.

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TABLE XII. R_{ON} (V_{DD}-V_{SS} = 4.5 V) check test - Continued.

Test condition	Parameter	Channel ON switch	Address switch				Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			S9	S10	S11	S12	A to pin	B to pin	Min	Max		
K25	R_{ON} (V_{DD}-V_{SS} = 4.5 V) check 1/	S1	L	H	H	L	24	27	1	70	mV	13
K26	"	S2	"	"	"	"	28	"	"	"	"	"
K27	"	S3	"	"	"	"	35	"	"	"	"	"
K28	"	S4	"	"	"	"	39	"	"	"	"	"
K29	"	S1	H	"	"	"	5	"	"	"	"	"
K30	"	S2	"	"	"	"	8	"	"	"	"	"
K31	"	S3	"	"	"	"	34	"	"	"	"	"
K32	"	S4	"	"	"	"	19	"	"	"	"	"
K33	"	S1	L	L	L	H	4	29	"	"	"	"
K34	"	S2	"	"	"	"	9	"	"	"	"	"
K35	"	S3	"	"	"	"	15	"	"	"	"	"
K36	"	S4	"	"	"	"	20	"	"	"	"	"
K37	"	S1	H	"	"	"	23	"	"	"	"	"
K38	"	S2	"	"	"	"	25	"	"	"	"	"
K39	"	S3	"	"	"	"	33	"	"	"	"	"
K40	"	S4	H	"	"	"	16	"	"	"	"	"
K41	"	S1	L	H	"	"	24	"	"	"	"	"
K42	"	S2	"	"	"	"	28	"	"	"	"	"
K43	"	S3	"	"	"	"	35	"	"	"	"	"
K44	"	S4	"	"	"	"	39	"	"	"	"	"
K45	"	S1	H	"	"	"	5	"	"	"	"	"
K46	"	S2	H	"	"	"	8	"	"	"	"	"

See footnote at end of table.

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TABLE XII. R_{ON} (V_{DD}-V_{SS} = 4.5 V) check test - Continued.

Test condition	Parameter	Channel ON switch	Address switch				Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			S9	S10	S11	S12	A to pin	B to pin	Min	Max		
K47	R_{ON} (V_{DD} - V_{SS} = 4.5 V) check 1/	S3	H	H	L	H	34	29	1	70	mV	13
K48	"	S4	H	H	L	"	19	29	"	"	"	"
K49	"	S1	L	L	H	"	4	18	"	"	"	"
K50	"	S2	"	"	"	"	9	"	"	"	"	"
K51	"	S3	"	"	"	"	15	"	"	"	"	"
K52	"	S4	"	"	"	"	20	"	"	"	"	"
K53	"	S1	H	"	"	"	23	"	"	"	"	"
K54	"	S2	"	"	"	"	25	"	"	"	"	"
K55	"	S3	"	"	"	"	33	"	"	"	"	"
K56	"	S4	"	"	"	"	16	"	"	"	"	"
K57	"	S1	L	H	"	"	24	"	"	"	"	"
K58	"	S2	"	"	"	"	28	"	"	"	"	"
K59	"	S3	"	"	"	"	35	"	"	"	"	"
K60	"	S4	"	"	"	"	39	"	"	"	"	"
K61	"	S1	H	"	"	"	5	"	"	"	"	"
K62	"	S2	"	"	"	"	8	"	"	"	"	"
K63	"	S3	"	"	"	"	34	"	"	"	"	"
K64	"	S4	"	"	"	"	19	"	"	"	"	"

1/ R_{ON} resistance (table 1) = $\frac{\text{Test condition K}}{1 \times 10^{-4}\text{A}}$ = Ω.

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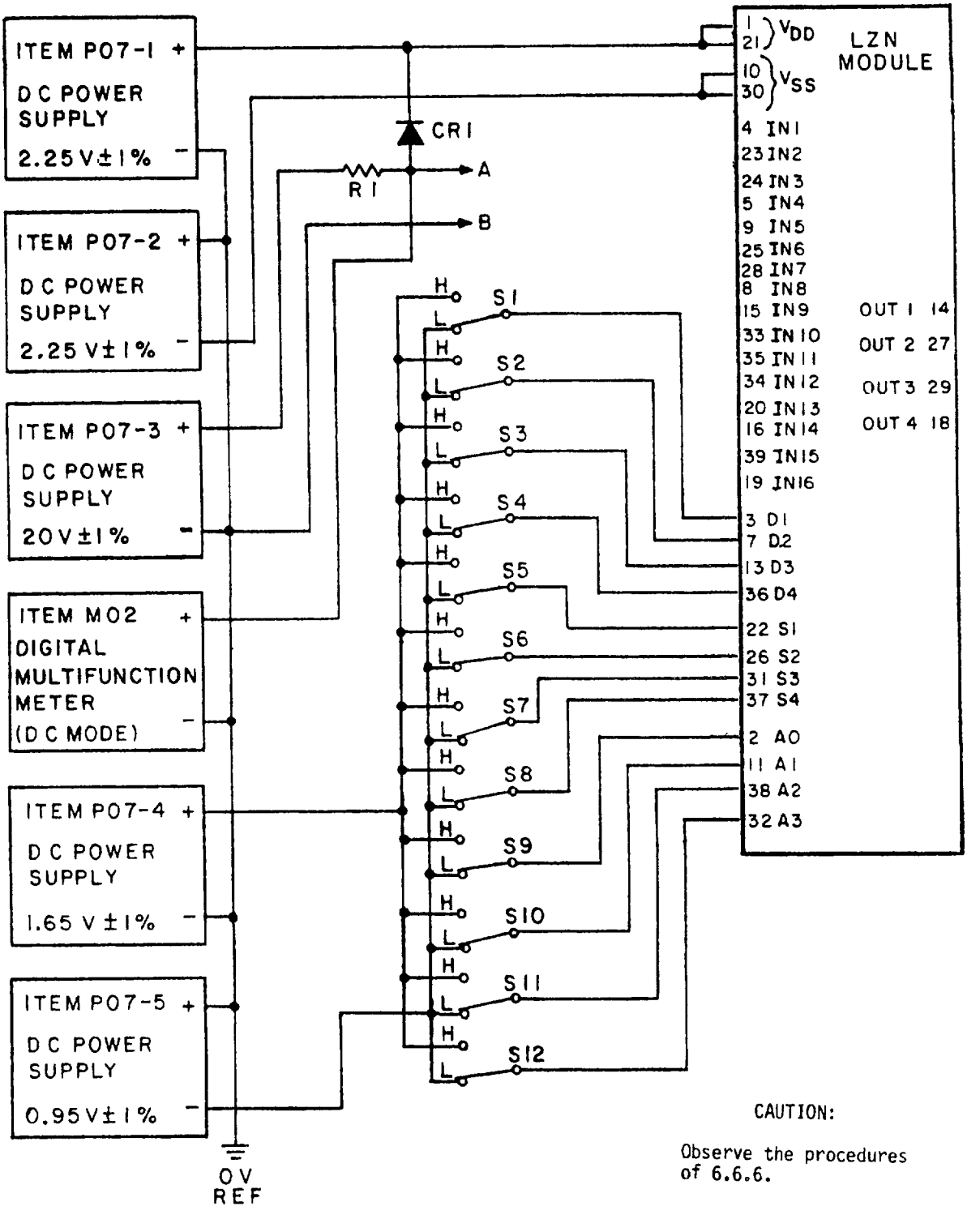


FIGURE 13. R_{ON} (V_{DD} - V_{SS} = 4.5 V) check test circuit for table XII.

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NOTES
1. R1 = 200 kΩ ±1%, 1/8 W.
2. CR1 = 1N4148, or equivalent.
3. S1-S12 = SPDT switch.
4. When power is initially applied, the analog switches may be either OFF or ON. To set all switches OFF perform the procedure of figure 14 (set logic high level = 1.65 V, logic low level = 0.95 V, and $V_{DD} = V_{SS} = 2.25$ V). This procedure shall be performed prior to setup per this figure.
5. The R_{ON} test shall be performed as follows
   a. Do not de-energize items P07-1, P07-2, P07-4 or P07-5 while changing from procedure of note 4 or during the remainder of the tests (if these are de-energized the procedure of note 4 must be repeated).
   b. Keep switches S1-S4 in position L (except as noted) and switch S5-S8 to H.
   c. Connect points A and B per table XII.
   d. Switch S9-S12 to the correct position per table XII.
   e. Switch the channel ON switch specified in table XII from L to H (this will turn the channel on and item M02 will change from a reading of 2.7 ±0.4 V (for reference only) to a within limit reading.
   f. Read item M02.
   g. Switch the channel ON switch from H to L (this will turn the channel off).
   h. Repeat steps c through g for each test condition of table XII.

FIGURE 13 R_{ON} ($V_{DD} - V_{SS} = 4.5$ V) check test circuit for table XII - Continued

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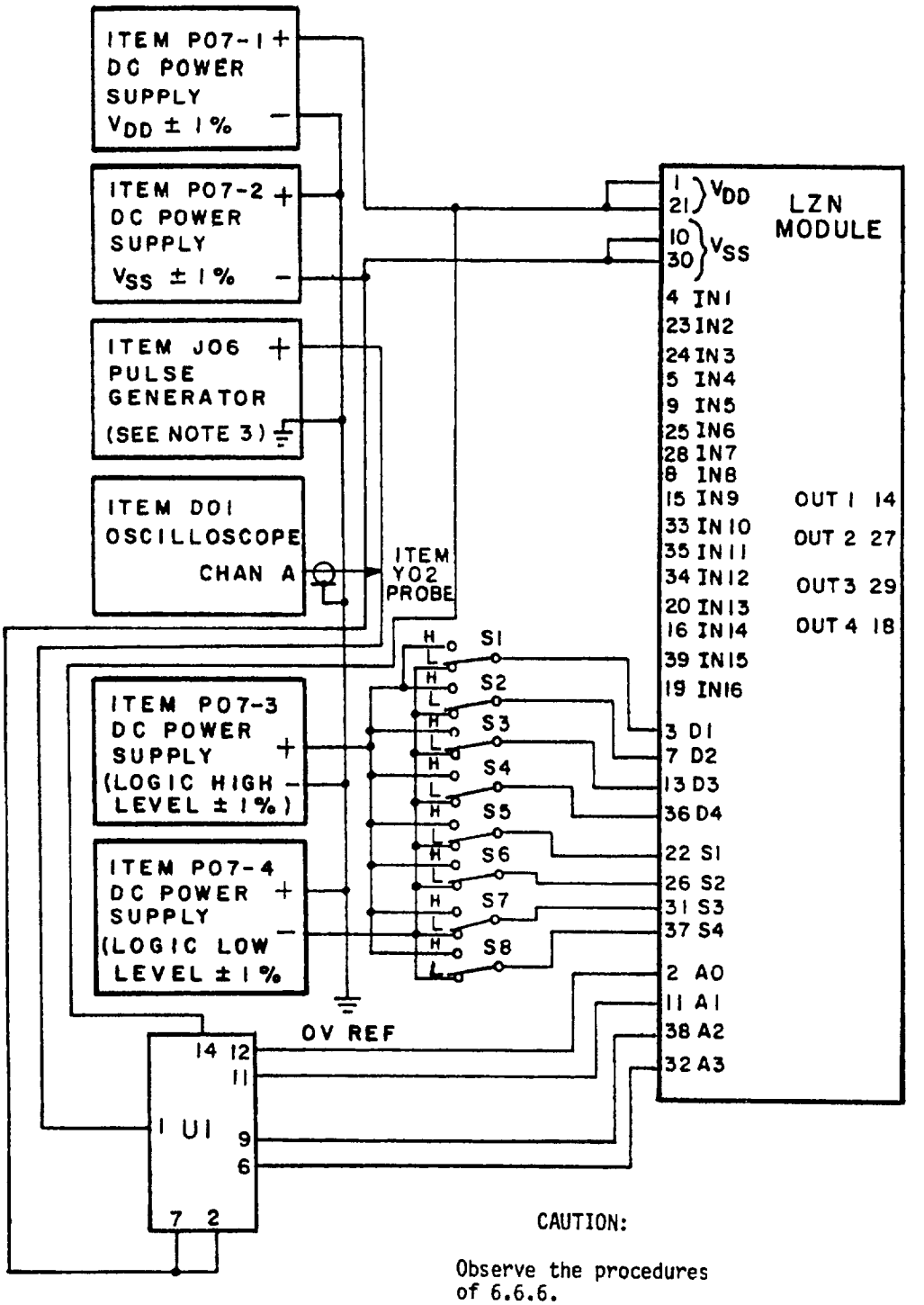


FIGURE 14. Test circuit for turning all analog switches to the OFF state (dual power supplies).

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NOTES:
1. S1-S8 = SPDT switch.
2. U1 = CD4022B, or equivalent.
3. Item J06 pulse parameters:
   PRP = 4 µs.
   PW = 2 µs
   $t_{TLH} = t_{THL}$ = 50 ±40 ns.
   Pulse high level = Same as V_{DD} +0%, -10% (module will be damaged if pulse high level is greater than V_{DD}).
   Pulse low level = Same as V_{SS} +10%, -0% (module will be damaged if pulse low level is less than V_{SS}).
4. Procedure

   a. Set switches S1 through S4 to position L.
   b. Set switches S5 through S8 to position H.
   c. Turn on items P07-1 and P07-2, then connect item J06 and turn on items P07-3 and P07-4. This will turn all analog switches off after U1 counts through all possible states (this will take approximately 64 µs).
   d. Set switches S5 through S8 to position L (this will disconnect the address pins 2, 11, 32, and 38 from the analog switches while disconnecting U1).
   e. Return to the applicable test condition while keeping items P07-1 and P07-2 energized.

FIGURE 14. Test circuit for turning all analog switches to the OFF state (dual power supplies) - Continued.

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* TABLE XIII. R_{ON} (variable input voltage) check test.

Test condition	Parameter 1/	Item P07-1 voltage V±1%	Item P07-2 voltage V±1%	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	Min	Max		
L1	R_{ON} (V_{IN}-V_{SS} = 0.2 V) variable input voltage check	4.3	0.2	4	14	1	70	mV	15
L2	"	"	"	9	"	"	"	"	"
L3	"	"	"	15	"	"	"	"	"
L4	"	"	"	20	"	"	"	"	"
L5	R_{ON} (V_{IN}-V_{SS} = 0.7 V) variable input voltage check	3.8	0.7	4	"	"	"	"	"
L6	"	"	"	9	"	"	"	"	"
L7	"	"	"	15	"	"	"	"	"
L8	"	"	"	20	"	"	"	"	"
L9	R_{ON} (V_{IN}-V_{SS} = 1.2 V) variable input voltage check	3.3	1.2	4	"	"	"	"	"
L10	"	"	"	9	"	"	"	"	"
L11	"	"	"	15	"	"	"	"	"
L12	"	"	"	20	"	"	"	"	"
L13	R_{ON} (V_{IN}-V_{SS} = 1.7 V) variable input voltage check	2.8	1.7	4	"	"	"	"	"
L14	"	"	"	9	"	"	"	"	"
L15	"	"	"	15	"	"	"	"	"
L16	"	"	"	20	"	"	"	"	"
L17	R_{ON} (V_{IN}-V_{SS} = 2.8 V) variable input voltage check	1.7	2.8	4	"	"	"	"	"
L18	"	"	"	9	"	"	"	"	"
L19	"	"	"	15	"	"	"	"	"
L20	"	"	"	20	"	"	"	"	"

See footnote at end of table.

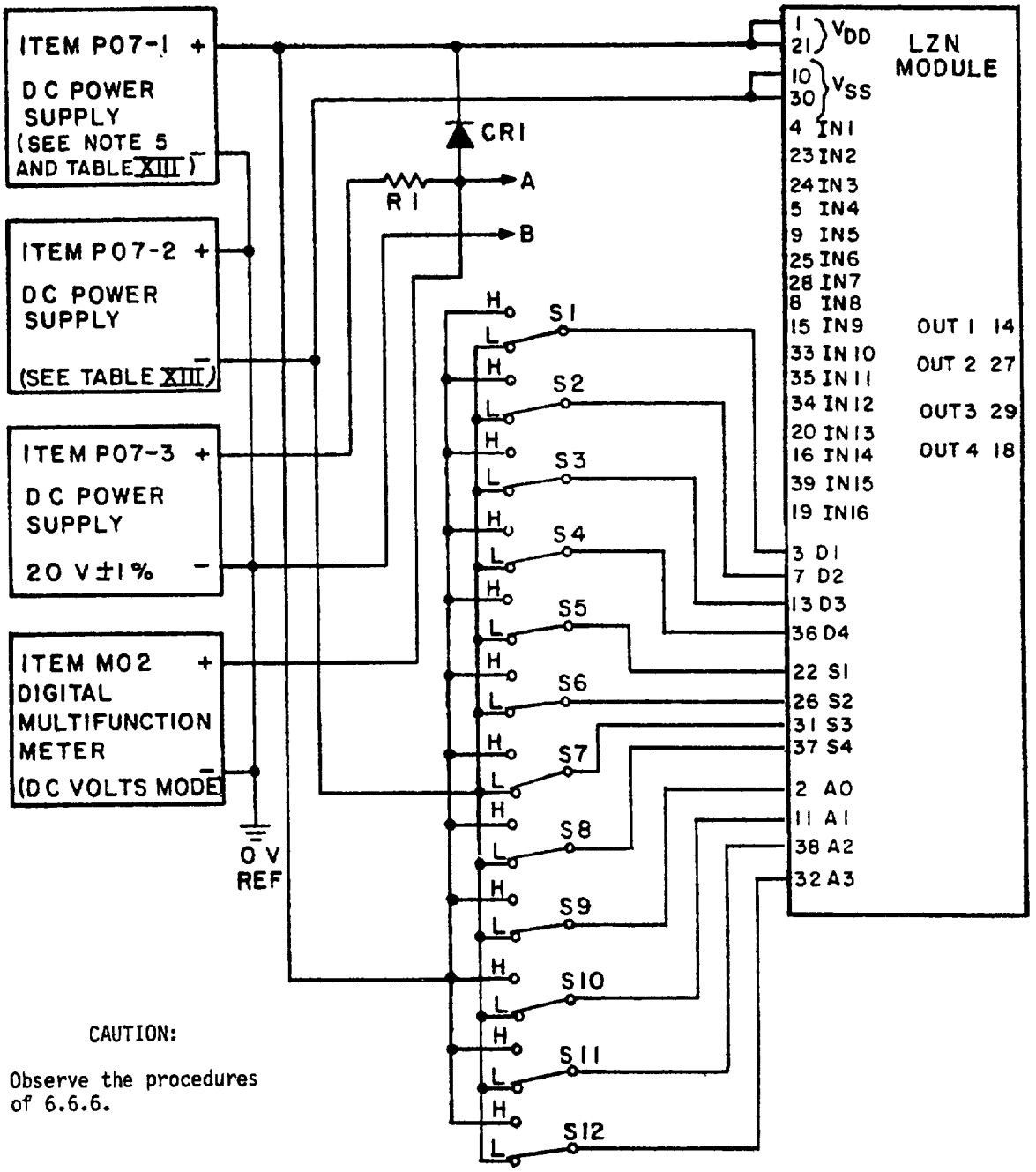
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* TABLE XIII. R_{ON} (variable input voltage) check test - Continued.

Test condition	Parameter 1/	Item P07-1 voltage V±1%	Item P07-2 voltage V±1%	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	Min	Max		
L21	R_{ON} (V_{IN}-V_{SS} = 3.3 V) variable input voltage check	1.2	3.3	4	14	1	70	mV	15
L22	"	"	"	9	"	"	"	"	"
L23	"	"	"	15	"	"	"	"	"
L24	"	"	"	20	"	"	"	"	"
L25	R_{ON} (V_{IN}-V_{SS} = 3.8 V) variable input voltage check	0.7	3.8	4	"	"	"	"	"
L26	"	"	"	9	"	"	"	"	"
L27	"	"	"	15	"	"	"	"	"
L28	"	"	"	20	"	"	"	"	"
L29	R_{ON} (V_{IN}-V_{SS} = 4.3 V) variable input voltage check	0.2	4.3	4	"	"	"	"	"
L30	"	"	"	9	"	"	"	"	"
L31	"	"	"	15	"	"	"	"	"
L32	"	"	"	20	"	"	"	"	"

1/ R_{ON} resistance (table I) = $\frac{\text{Test condition L}}{1 \times 10^{-4}\text{A}}$ = Ω.

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* FIGURE 15. R_{ON} (variable input voltage) check test circuit for table XIII.

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NOTES:
1. R1 = 200 kΩ ±1%, 1/8 W.
2. S1-S12 = SPDT switch.
3. CR1 = 1N4148, or equivalent.
4. When power is initially applied, the analog switches may be either OFF or ON. To set all switches OFF perform the procedure of figure 14 (delete power supplies P07-3 and P07-4 on figure 14, connect H of switches S1-S8 to pin 21, L to pin 30, set V_{DD} = 4.3 V, and V_{SS} = 0.2 V).

   Item P07-1 must sink current during some of the tests. Use a power supply capable of sinking current or add an external resistor between the plus and minus terminals.
6. The tests shall be performed as follows:

   a. Place switches S5-S8 (strobe) to position L.
   b. Place switches S1-S4 (data-in) to position H.
   c. Place switches S9-S12 to position L.
   d. Switch S5-S8 (one at a time) from L to H and back to L (this turns the four analog switches to be tested to the ON state).
   e. Without de-energizing item P07-1 and P07-2 make connections and voltage changes per table XIII and read the item M02 voltage for each test condition.

* FIGURE 15. R_{ON} (variable input voltage) check test circuit for table XIII - Continued.

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* TABLE XIV. AC feedthrough check test.

Test condition	Parameter		Connect B to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
M1	OUT1	AC feedthrough check	14	0.000	0.092	V rms	16
M2	OUT2	"	27	"	"	"	"
M3	OUT3	"	29	"	"	"	"
M4	OUT4	"	18	"	"	"	"

* TABLE XV. Large signal response test. 1/

Test condition	Parameter		Channel ON switch	Address switch S9	S10	S11	S12	Connect A to pin	B to pin	Initial limits 25°C ±5°C ambient Min	Max	Unit	Test figure
N1	OUT1	Large signal response	S1	H	H	L	L	18,27,29	14	4.4	4.7	V rms	16
N2	OUT2	"	S2	H	H	H	L	14,18,29	27	"	"	"	"
N3	OUT3	"	S3	H	H	L	H	14,18,27	29	"	"	"	"
N4	OUT4	"	S4	H	H	H	H	14,27,29	18	"	"	"	"

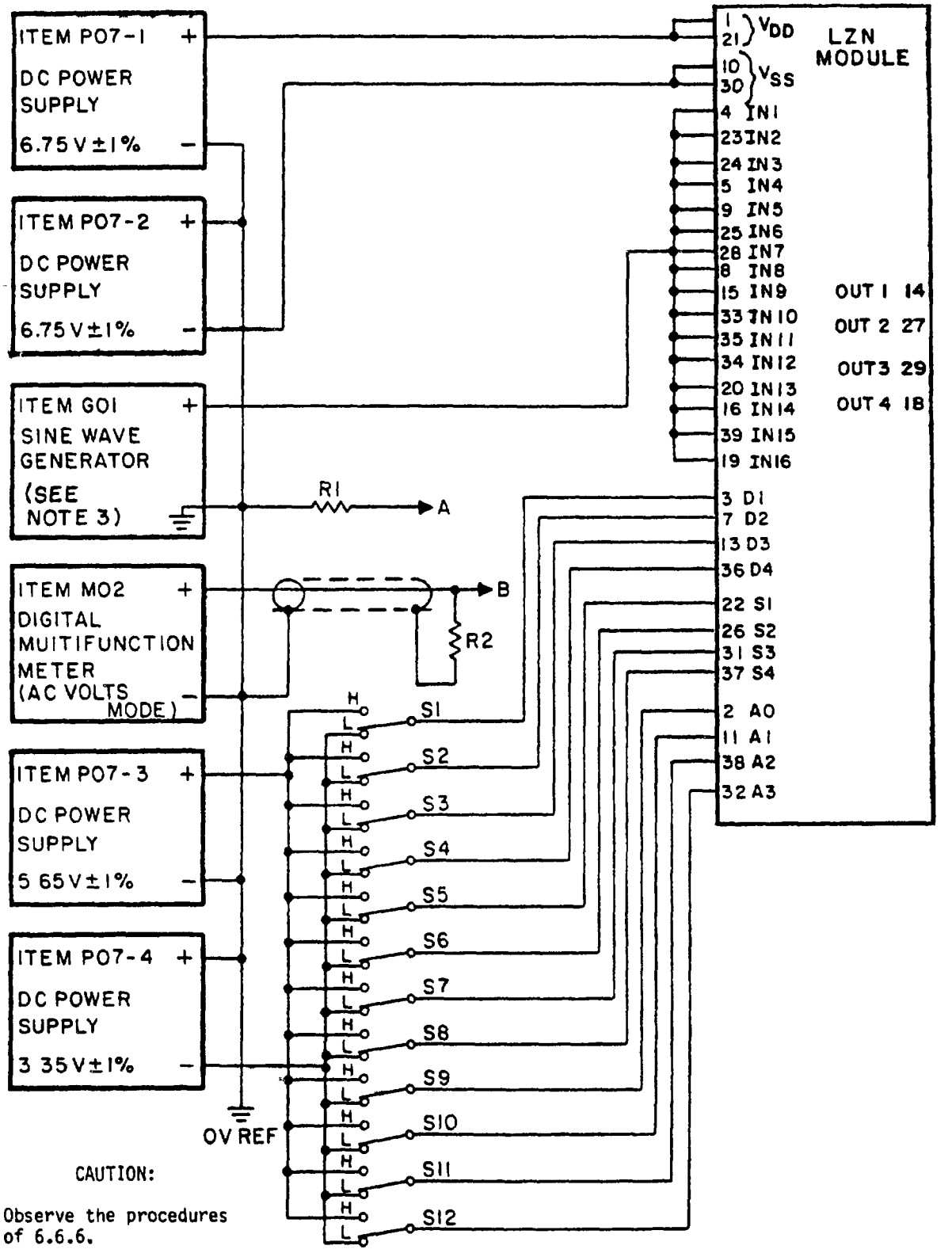
1/ This test can be performed concurrently with the output crosstalk test.

* TABLE XVI. Output crosstalk check test. 1/

Test condition	Parameter		Channel ON switch	Address switch S9	S10	S11	S12	Connect A to pin	B to pin	Initial limits 25°C ±5°C ambient Min	Max	Unit	Test figure
P1	OUT1	Output cross-talk check	S1	H	H	L	L	14	18,27,29	0.00	0.28	V rms	16
P2	OUT2	"	S2	H	H	H	L	27	14,18,29	"	"	"	"
P3	OUT3	"	S3	H	H	L	H	29	14,18,27	"	"	"	"
P4	OUT4	"	S4	H	H	H	H	18	14,27,29	"	"	"	"

1/ This test can be performed concurrently with the large signal response test.

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* FIGURE 16. AC feedthrough check, large signal response, and output crosstalk check test circuit for tables XIV, XV, and XVI.

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NOTES:
1. R1 = R2 = 10 kΩ ±1%, 1/8 W.
2. S1 - S12 = SPDT switch.
3. Item G01 setup:
   Amplitude = 4.6 V rms ±1% (recheck before each test condition).
   Frequency = 20 kHz ±1%.
   DC offset = ±100 mV maximum.
4. When input power is initially applied the analog switches may be either OFF or ON. To set all switches OFF perform the procedure per figure 14 (set logic high level = 5.56 V, logic low level = 3.35 V, and $V_{DD} = V_{SS} = 6.75$ V). This procedure shall be performed prior to setup per this figure.
5. Do not de-energize items P07-1 through P07-4 while changing from the procedure of note 4 or during the reminder of the test of this figure (if these are de-energized the procedure of note 4 must be repeated).
6. The AC feedthrough test (test conditions M1-M4) shall be performed as follows:

   Connect point B per table XIV and read item M02 for each test condition (point A is open for these test conditions).
   Keep switches S1-S12 in position L.

7. The large signal response test (test conditions N1-N4) and/or the output crosstalk test (test conditions P1-P4) shall be performed as follows (the only difference in these tests is the pins to which point A and B are connected and the test limits):

   a. Keep switches S1-S4 in position L (except as noted) and switch S5-S8 to position H.
   b. Switch S9-S12 to the position indicated in table XV or XVI.
   c. Connect points A and B and switch the channel ON switch specified in table XV or XVI from position L to H. (This will turn one analog channel on.)
   d. Read item M02.
   e. Switch the channel ON switch from position H to L.
   f. Repeat steps b through e for each test condition.

* FIGURE 16. AC feedthrough check, large signal response, and output crosstalk check test circuit for tables XIV, XV, and XVI - Continued.

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* TABLE XVII. Input crosstalk check test.

Test condition	Parameter	Channel ON switch	Address switch				Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			S9	S10	S11	S12	A to pin	B to pin	Min	Max		
Q1	Input crosstalk check	S1	L	H	L	L	24	4,5,23, 25	0.00	0.28	V rms	17
Q2	"	S2	L	H	H	L	28	8,9	"	"	"	"
Q3	"	S3	H	H	L	H	34	15,16, 33,35	"	"	"	"
Q4	"	S4	L	H	H	H	39	19,20	"	"	"	"

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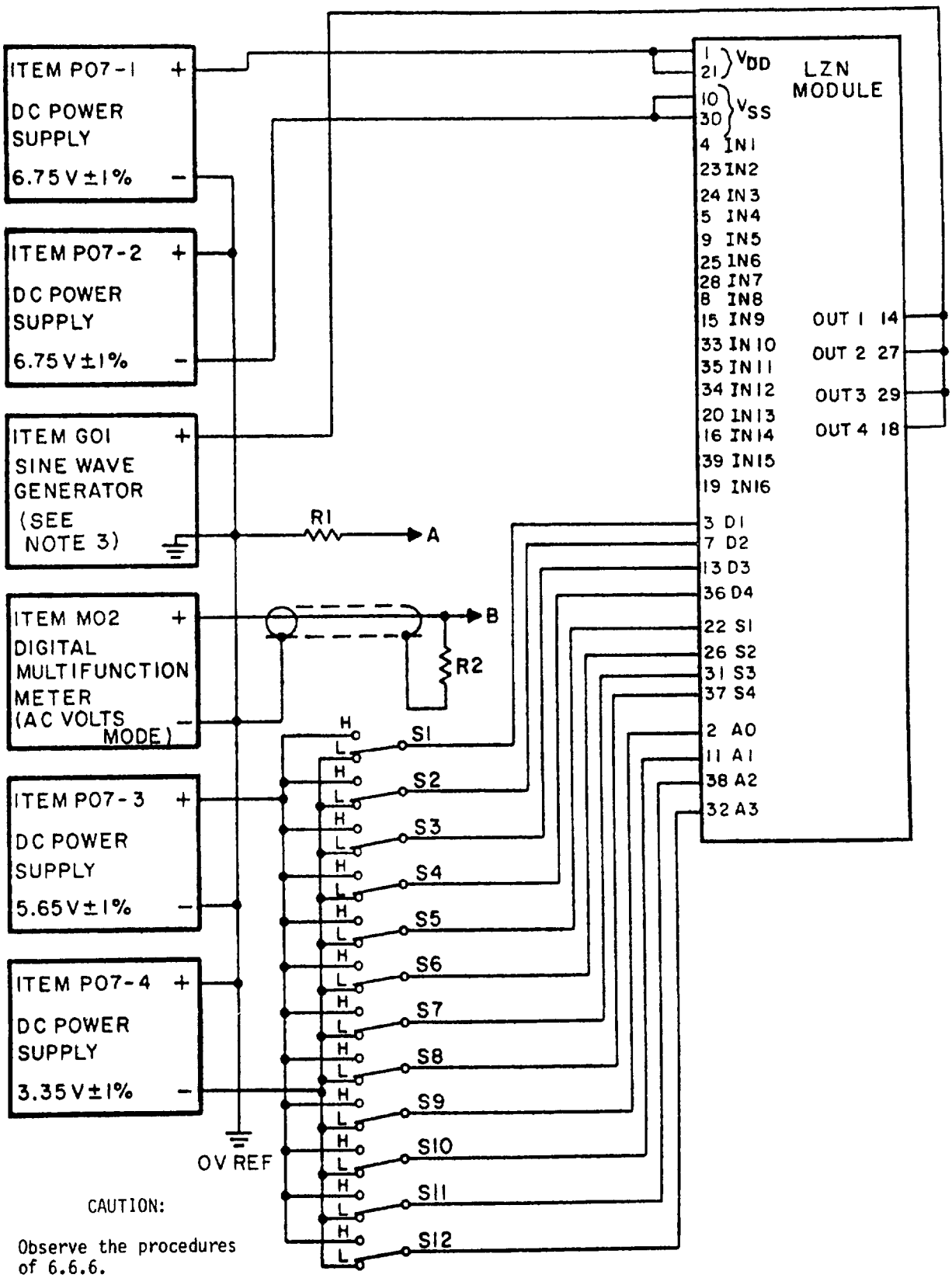


FIGURE 17. Input crosstalk check test circuit for table XVII.

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NOTES:
1. R1 = R2 = 10 kΩ ±1%, 1/8 W.
2. S1-S12 = SPDT switch.
3. Item G01 setup:
   Amplitude = 4.6 V rms ±1%.
   Frequency = 20 kHz ±1%.
   DC offset = ±100 mV maximum.
4. When input power is initially applied the analog switches may be either OFF or ON. To set all switches OFF perform the procedure of figure 14 (set logic high level = 5.65 V, logic low level = 3.35 V, and $V_{DD} = V_{SS} = 6.75$ V). This procedure shall be performed prior to setup per this figure.
5. The input crosstalk test shall be performed as follows:

   a. Do not de-energize items P07-1 through P07-4 while changing from the procedure of note 4 or during the remainder of the tests (if these are de-energized the procedure of note 4 must be repeated).
   b. Keep switches S1-S4 in position L (except as noted) and switch S5-S8 to position H.
   c. Switch S9-S12 to the positions indicated in table XVII.
   d. Connect points A and B and switch the channel ON switch specified in table XVII for position L to H (this will turn one analog switch on).
   e. Read item M02.
   f. Switch the channel ON switch from position H to L.
   g. Repeat steps b through f for each test condition.

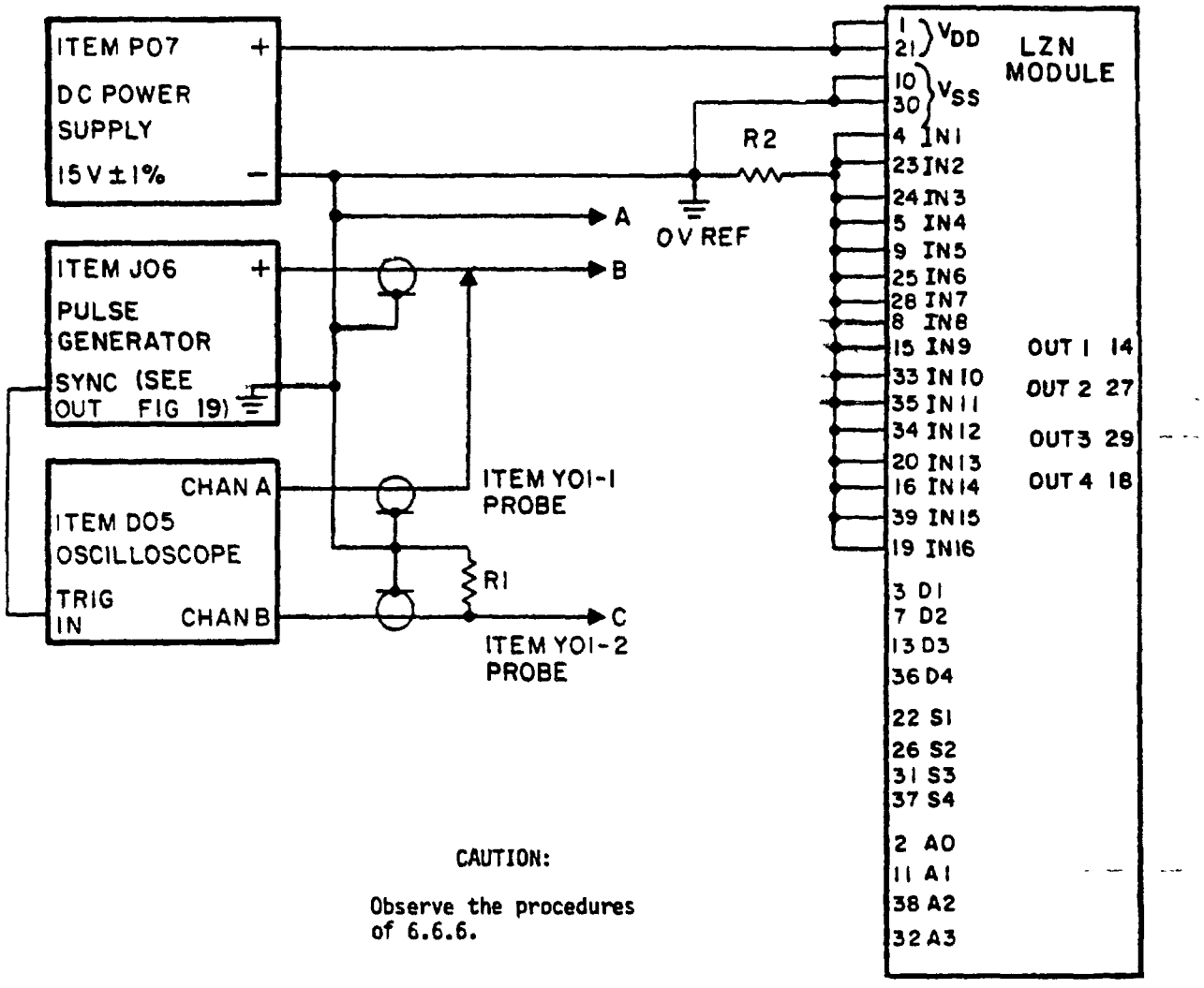
FIGURE 17. Input crosstalk check test circuit for table XVII - Continued.

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* TABLE XVIII. Control crosstalk (address, data-in, and strobe) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	Control crosstalk, address	3,7,13,22,26, 31,36,37	2,11,32,38	14	-450	450	mV	18,19
R2	"	"	"	18	"	"	"	"
R3	"	"	"	27	"	"	"	"
R4	"	"	"	29	"	"	"	"
R5	Control crosstalk, data-in	2,11,22,26,31, 32,37,38	3,7,13,36	14	"	"	"	"
R6	"	"	"	18	"	"	"	"
R7	"	"	"	27	"	"	"	"
R8	"	"	"	29	"	"	"	"
R9	Control crosstalk, strobe	2,3,7,11,13,32, 36,38	22,26,31,37	14	"	"	"	"
R10	"	"	"	18	"	"	"	"
R11	"	"	"	27	"	"	"	"
R12	"	"	"	29	"	"	"	"

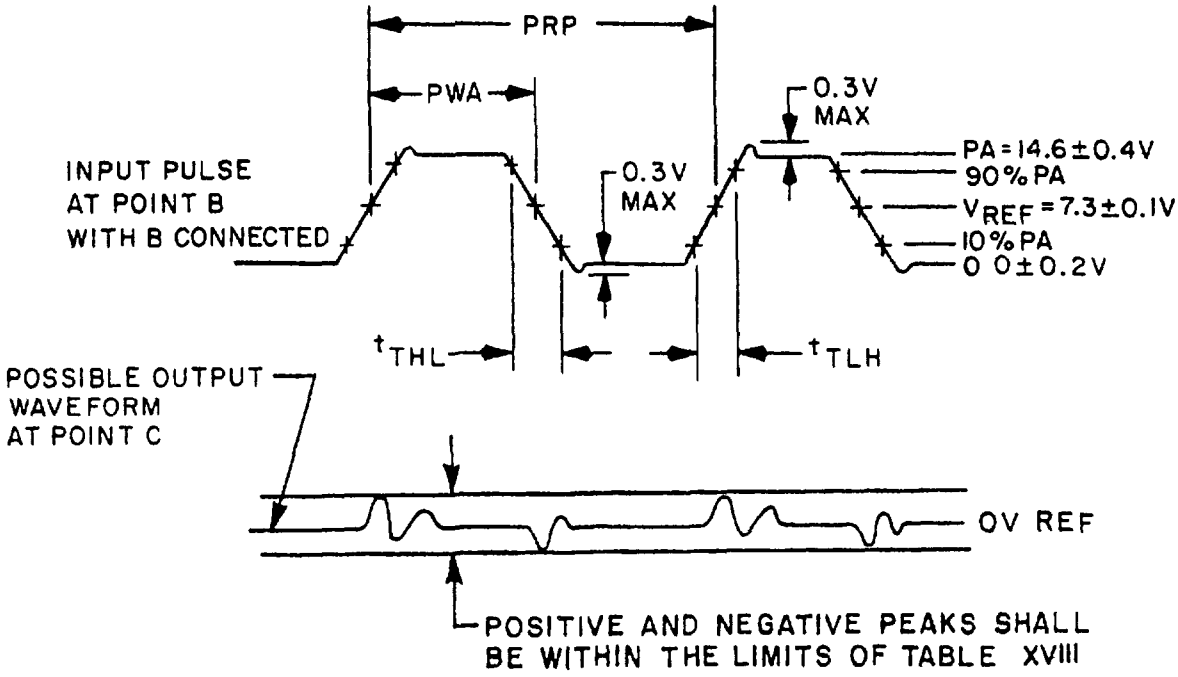
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NOTES:
1. R1 = 10 kΩ ±1%, 1/8 W.
2. R2 = 1 kΩ ±1%, 1/8 W.
3. When input power is initially applied the analog switches may be either OFF or ON. To set all switches OFF perform the procedure of figure 6 (set V_{DD} = 15 V, logic high level = 15 V (use V_{DD} power supply), and logic low level = 0 V (no power supply required).
4. Do not de-energize item P07 when changing from the procedure of note 3 or during the remainder of the tests.
5. When making connections per table XVIII always connect point A before connecting point B. Always disconnect point B before disconnecting point A. If this sequence is not followed some analog switches will accidently be turned on. Point C can be connected in any sequence.

* FIGURE 18. Control crosstalk (address, data-in, and strobe) test circuit for table XVIII.

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NOTE. Input pulse parameters:
$t_{TLH} = t_{THL} = 15 \pm 2$ ns.
PWA = 2.5 ±0.3 µs.
PRP = 5.0 ±0.5 µs.

* FIGURE 19. Control crosstalk (address, data-in, and strobe) test circuit waveforms for table XVIII.

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* TABLE XIX. *Input current (high-level) test.*

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
S1	D1 Input current, high level, data	2,7,11,13,22,26,31,32,36,37,38	3	0	200	nA	20
S2	D2 "	2,3,11,13,22,26,31,32,36,37,38	7	"	"	"	"
S3	D3 "	2,3,7,11,22,26,31,32,36,37,38	13	"	"	"	"
S4	D4 "	2,3,7,11,13,22,26,31,32,37,38	36	"	"	"	"
S5	S1 Input current, high level, strobe	2,3,7,11,13,26,31,32,36,37,38	22	"	"	"	"
S6	S2 "	2,3,7,11,13,22,31,32,36,37,38	26	"	"	"	"
S7	S3 "	2,3,7,11,13,22,26,32,36,37,38	31	"	"	"	"
S8	S4 "	2,3,7,11,13,22,26,31,32,36,38	37	"	"	"	"
S9	A0 Input current, high level, address	3,7,11,13,22,26,31,32,36,37,38	2	"	"	"	"
S10	A1 "	2,3,7,13,22,26,31,32,36,37,38	11	"	"	"	"
S11	A2 "	2,3,7,11,13,22,26,31,32,36,37	38	"	"	"	"
S12	A3 "	2,3,7,11,13,22,26,31,36,37,38	32	"	"	"	"

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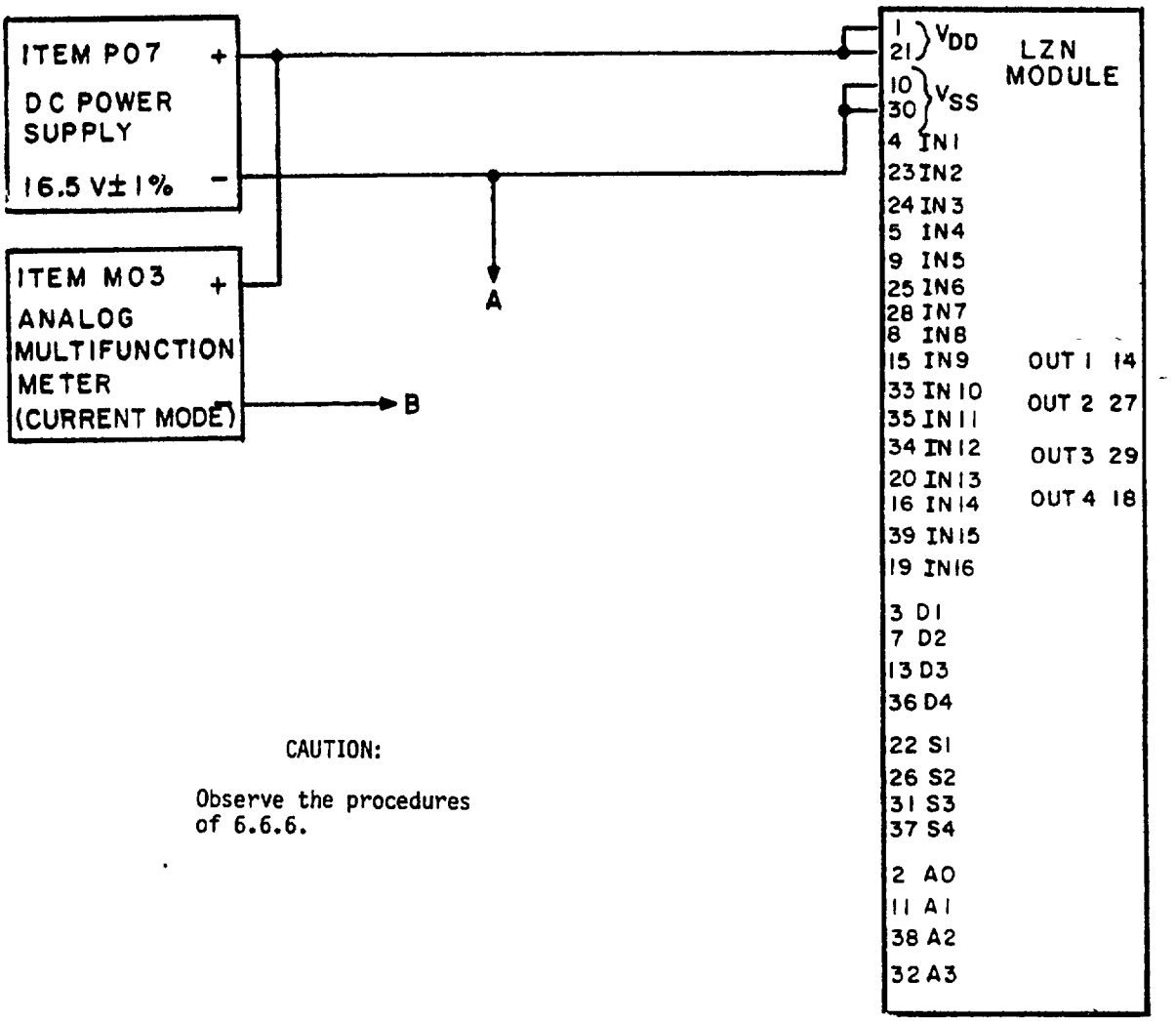


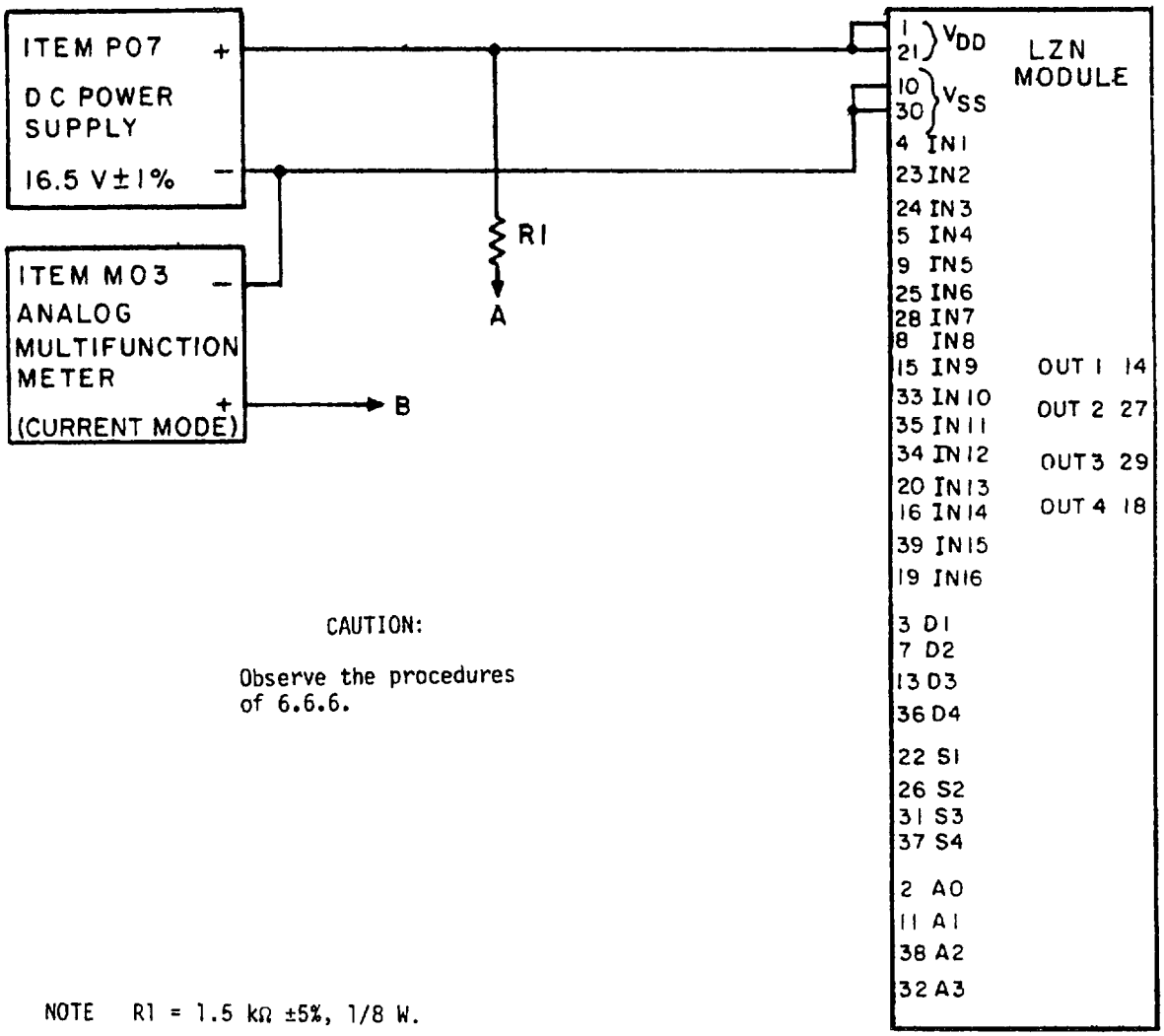
FIGURE 20. Input current (high-level) test circuit for table XIX.

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* TABLE XX. Input current (low-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T1	D1 Input current, low level, data	2,7,11,13,22,26,31,32,36,37,38	3	0	200	nA	21
T2	D2 "	2,3,11,13,22,26,31,32,36,37,38	7	"	"	"	"
T3	D3 "	2,3,7,11,22,26,31,32,36,37,38	13	"	"	"	"
T4	D4 "	2,3,7,11,13,22,26,31,32,37,38	36	"	"	"	"
T5	S1 Input current, low level, strobe	2,3,7,11,13,26,31,32,36,37,38	22	"	"	"	"
T6	S2 "	2,3,7,11,13,22,31,32,36,37,38	26	"	"	"	"
T7	S3 "	2,3,7,11,13,22,26,32,36,37,38	31	"	"	"	"
T8	S4 "	2,3,7,11,13,22,26,31,32,36,38	37	"	"	"	"
T9	A0 Input current, low level, address	3,7,11,13,22,26,31,32,36,37,38	2	"	"	"	"
T10	A1 "	2,3,7,13,22,26,31,32,36,37,38	11	"	"	"	"
T11	A2 "	2,3,7,11,13,22,26,31,32,36,37	38	"	"	"	"
T12	A3 "	2,3,7,11,13,22,26,31,36,37,38	32	"	"	"	"

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CAUTION:

Observe the procedures of 6.6.6.

NOTE R1 = 1.5 kΩ ±5%, 1/8 W.

FIGURE 21. Input current (low-level) test circuit for table XX.

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* TABLE XXI. Power supply filter capacitance test.

Test condition	Parameter		Connect Item C01 1/		Initial limits 25°C ±5°C ambient		Unit	Test figure
			+ terminal to pin	- terminal to pin	Min	Max		
U	Power supply filter	capacitance	1	10	0.075	0.125	µF	---

1/ Item C01 setup:
Frequency = 1 kHz ±1%.
V_{pk} = 0.3 V maximum.

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TABLE XXII. Input and output capacitance (digital and switch) test.

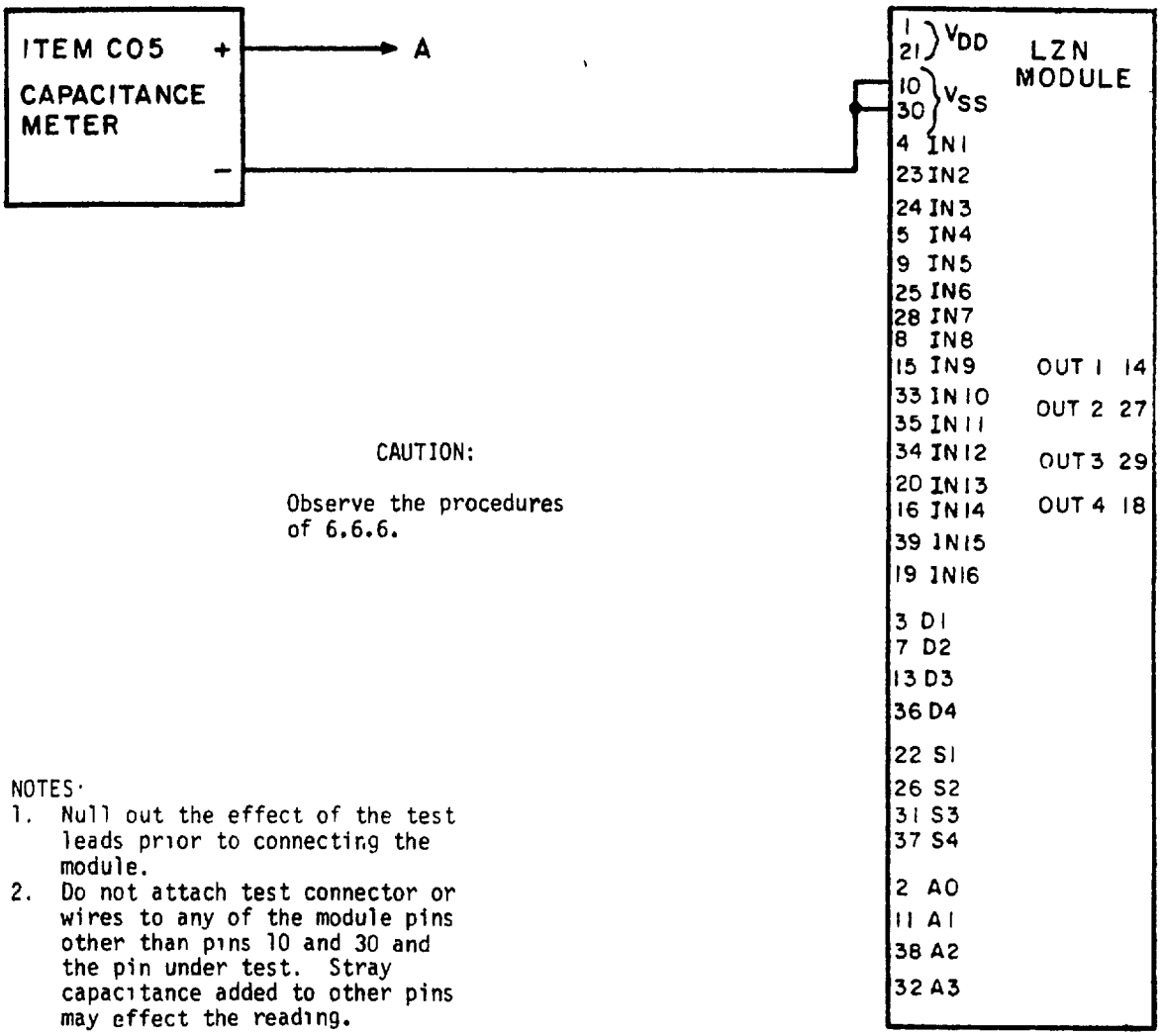
Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
V1	A0 Input capacitance, digital	2	2	25	pF	22
V2	D1 "	3	"	"	"	"
V3	D2 "	7	"	"	"	"
V4	A1 "	11	"	"	"	"
V5	D3 "	13	"	"	"	"
V6	S1 "	22	"	"	"	"
V7	S2 "	26	"	"	"	"
V8	S3 "	31	"	"	"	"
V9	A3 "	32	"	"	"	"
V10	D4 "	36	"	"	"	"
V11	S4 "	37	"	"	"	"
V12	A2 "	38	"	"	"	"
V13	IN1 Input capacitance, switch	4	"	60	"	"
V14	IN4 "	5	"	"	"	"
V15	IN8 "	8	"	"	"	"
V16	IN5 "	9	"	"	"	"
V17	IN9 "	15	"	"	"	"
V18	IN14 "	16	"	"	"	"
V19	IN16 "	19	"	"	"	"
V20	IN13 "	20	"	"	"	"

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* TABLE XXII. Input and output capacitance (digital and switch) test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
V21	IN2 Input capacitance, switch	23	2	60	pF	22
V22	IN3 "	24	"	"	"	"
V23	IN6 "	25	"	"	"	"
V24	IN7 "	28	"	"	"	"
V25	IN10 "	33	"	"	"	"
V26	IN12 "	34	"	"	"	"
V27	IN11 "	35	"	"	"	"
V28	IN15 "	39	"	"	"	"
V29	OUT1 Output capacitance, switch	14	10	300	"	"
V30	OUT4 "	18	"	"	"	"
V31	OUT2 "	27	"	"	"	"
V32	OUT3 "	29	"	"	"	"

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NOTES:
1. Null out the effect of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pins 10 and 30 and the pin under test. Stray capacitance added to other pins may effect the reading.

FIGURE 22. Input and output capacitance (digital and switch) test circuit for table XXII.

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* TABLE XXIII. Propagation delay time (address to output, t_{PHZ}) test. 1/

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W1	Propagation delay time, address to output (t_{PHZ}), V_{DD} = 5	Open	7,13,36	3	2,11,32, 38	14	100	1000	ns	23,24
W2	"	38	2,3,13, 36	7	11,32	27	"	"	"	"
W3	"	32	2,3,7,36	13	11,38	29	"	"	"	"
W4	"	32,38	3,7,13	36	2,11	18	"	"	"	"

1/ This test may be performed concurrently with test conditions X1-X4.

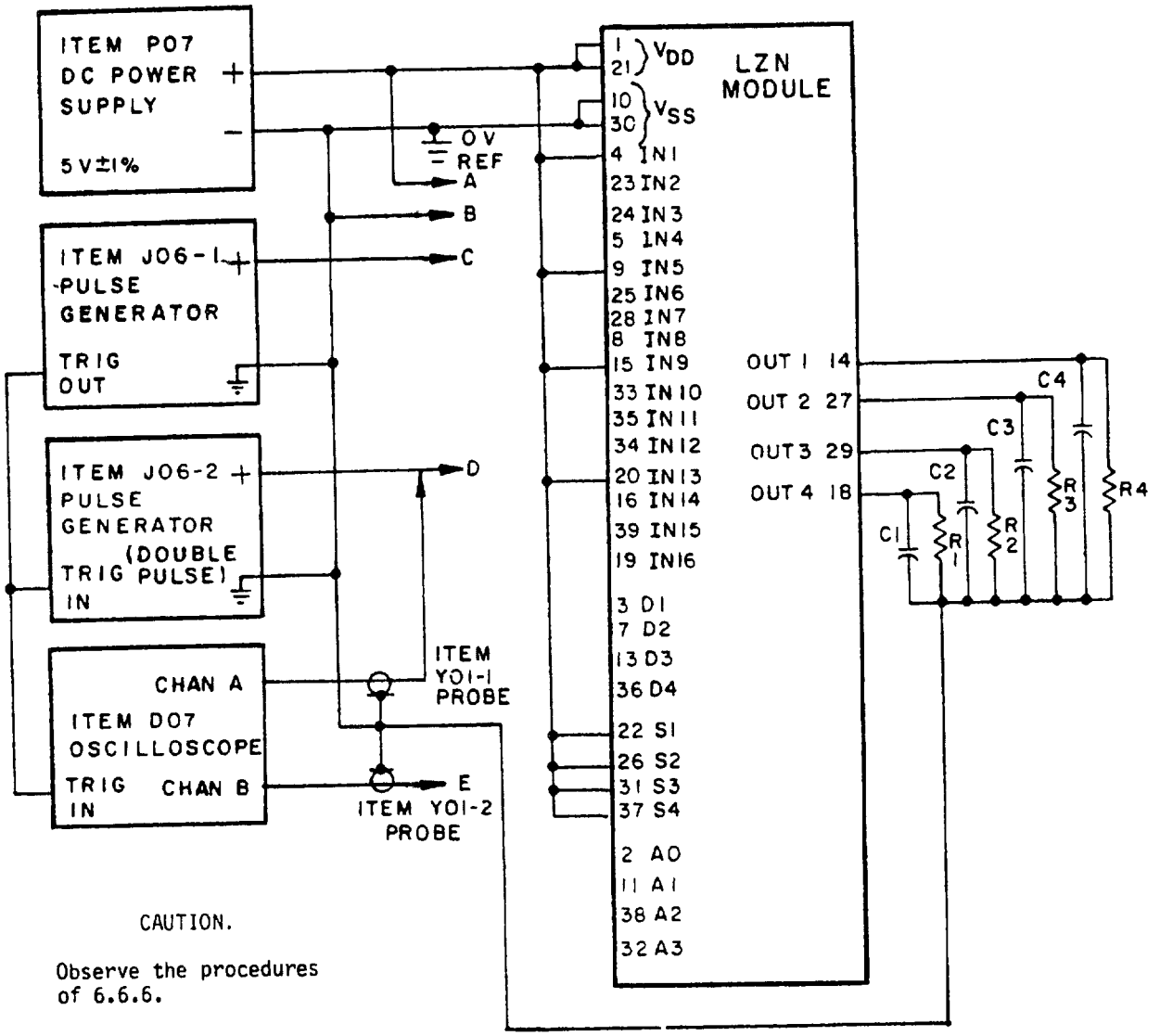
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* TABLE XXIV. Propagation delay time (address to output, t_{PZH}) test. 1/

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
X1	Propagation delay time, address to output (t_{PZH}), V_{DD} = 5 V	Open	7,13,36	3	2,11,32, 38	14	70	700	ns	23,24
X2	"	38	2,3,13, 36	7	11,32	27	"	"	"	"
X3	"	32	2,3,7,36	13	11,38	29	"	"	"	"
X4	"	32,38	3,7,13	36	2,11	18	"	"	"	"

1/ This test may be performed concurrently with test conditions W1-W4.

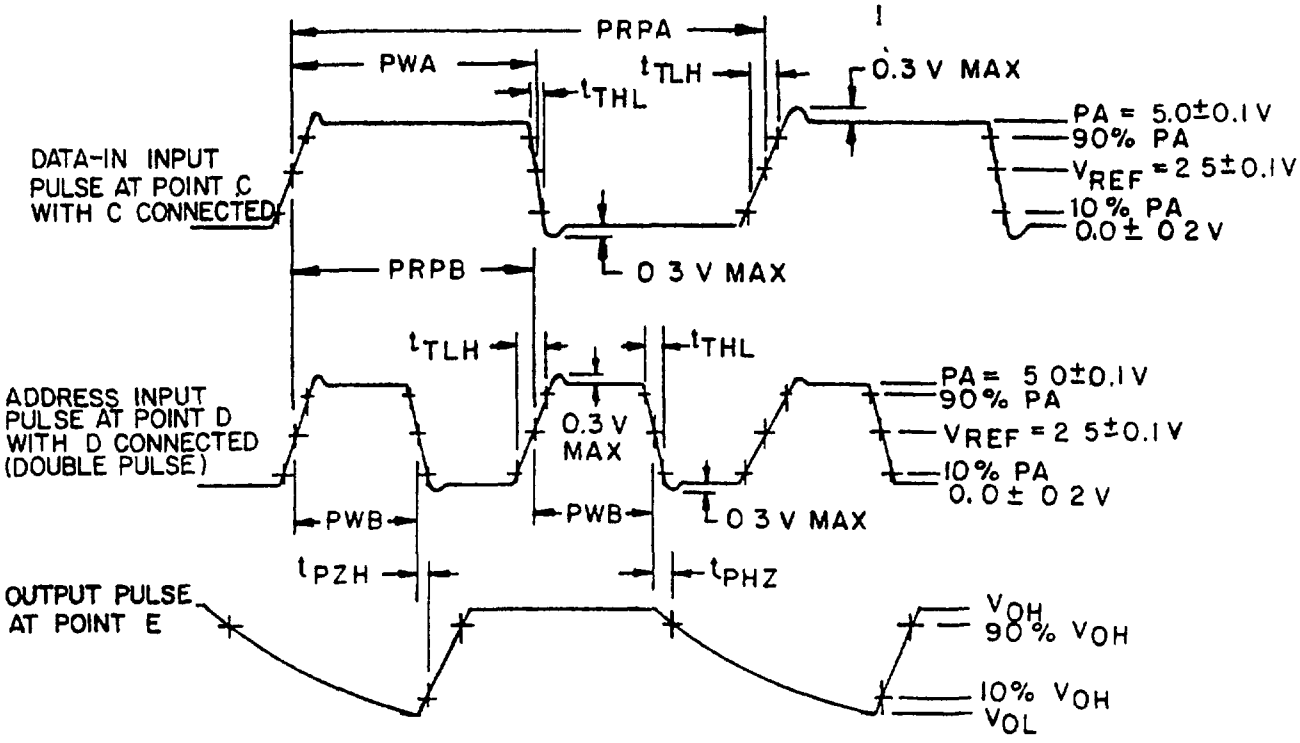
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NOTES:
1. R1-R4 = 10 kΩ ±1%, 1/8 W.
2. C1-C4 = 47 pF ±5% (including probe and parasitic capacitance).
3. For accurate test results each analog switch must initially be OFF. Set the switches to OFF prior to each test condition by performing the procedure of figure 6 (set V_{DD} = 5 V ±1%, logic high level = same as V_{DD} and logic low level = 0 V).

* FIGURE 23. Propagation delay time (address to output, t_{PHZ} and t_{PZH}) test circuit for tables XXIII and XXIV.

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NOTES:
1. Input pulse parameters:
   PRPA = 20 ±1 µs.
   PRPB = 10.0 ±0.5 µs.
   PWA = 11.0 ±0.5 µs.
   PWB = 5.0 ±0.3 µs.
   $t_{LTH} = t_{THL} = 20 \pm 4$ ns.
2. Before making the measurements specified in tables XXIII and XXIV, verify that the output pulse high level is $\geq$ 3.5 V and low level is $\leq$ 1.0 V.
3. The lead length of electrical conductors between resistors and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to 0 V reference except V_{OH} which is referenced to V_{OL}.

* FIGURE 24. Propagation delay time (address to output, t_{PHZ} and t_{PZH}) test circuit waveforms for tables XXIII and XXIV.

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* TABLE XXV. Propagation delay time (strobe to output, t_{PHZ}) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y1	Propagation delay time, strobe to output (t_{PHZ}), V_{DD} = 5 V	4	2,11,32,38,26,31,37	22	14	Open	70	700	ns	25,26
Y2	"	9	2,11,22,31,32,37,38	26	"	"	"	"	"	"
Y3	"	15	2,11,22,26,32,37,38	31	"	"	"	"	"	"
Y4	"	20	2,11,22,26,31,32,38	37	"	"	"	"	"	"
Y5	"	23,2	11,26,31,32,37,38	22	"	"	"	"	"	"
Y6	"	25,2	11,22,31,32,37,38	26	"	"	"	"	"	"
Y7	"	33,2	11,22,26,32,37,38	31	"	"	"	"	"	"
Y8	"	16,2	11,22,26,31,32,38	37	"	"	"	"	"	"
Y9	"	24,11	2,26,31,32,37,38	22	"	"	"	"	"	"
Y10	"	28,11	2,22,31,32,37,38	26	"	"	"	"	"	"
Y11	"	35,11	2,22,26,32,37,38	31	"	"	"	"	"	"
Y12	"	39,11	2,22,26,31,32,38	37	"	"	"	"	"	"
Y13	"	5,2,11	26,31,32,37,38	22	"	"	"	"	"	"
Y14	"	8,2,11	22,31,32,37,38	26	"	"	"	"	"	"
Y15	"	34,2,11	22,26,32,37,38	31	"	"	"	"	"	"
Y16	"	19,2,11	22,26,31,32,38	37	"	"	"	"	"	"
Y17	"	4,38	2,11,26,31,32,37	22	27	"	"	"	"	"
Y18	"	9,38	2,11,22,31,32,37	26	"	"	"	"	"	"
Y19	"	15,38	2,11,22,26,32,37	31	"	"	"	"	"	"
Y20	"	20,38	2,11,22,26,31,32	37	"	"	"	"	"	"

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Check the source to verify that this is the current version before use.

* TABLE XXV. Propagation delay time (strobe to output, t_{PHZ}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y21	Propagation delay time, strobe to output (t_{PHZ}), V_{DD} = 5 V	23,2, 38	11,26,31,32,37	22	27	Open	70	700	ns	25,26
Y22	"	25,2, 38	11,22,31,32,37	26	"	"	"	"	"	"
Y23	"	33,2, 38	11,22,26,32,37	31	"	"	"	"	"	"
Y24	"	16,2, 38	11,22,26,31,32	37	"	"	"	"	"	"
Y25	"	24,11, 38	2,26,31,32,37	22	"	"	"	"	"	"
Y26	"	28,11, 38	2,22,31,32,37	26	"	"	"	"	"	"
Y27	"	35,11, 38	2,22,26,32,37	31	"	"	"	"	"	"
Y28	"	39,11, 38	2,22,26,31,32	37	"	"	"	"	"	"
Y29	"	5,2, 11,38	26,31,32,37	22	"	"	"	"	"	"
Y30	"	8,2, 11,38	22,31,32,37	26	"	"	"	"	"	"
Y31	"	34,2, 11,38	22,26,32,37	31	"	"	"	"	"	"
Y32	"	19,2, 11,38	22,26,31,32	37	"	"	"	"	"	"
Y33	"	4,32	2,11,26,31,37,38	22	29	"	"	"	"	"
Y34	"	9,32	2,11,22,31,37,38	26	"	"	"	"	"	"
Y35	"	15,32	2,11,22,26,37,38	31	"	"	"	"	"	"
Y36	"	20,32	2,11,22,26,31,38	37	"	"	"	"	"	"
Y37	"	23,2, 32	11,26,31,37,38	22	"	"	"	"	"	"

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* TABLE XXV. Propagation delay time (strobe to output, t_{PHZ}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y38	Propagation delay time, strobe to output (t_{PHZ}), V_{DD} = 5 V	25,2, 32	11,22,31,37,38	26	29	Open	70	700	ns	25,26
Y39	"	33,2, 32	11,22,26,37,38	31	"	"	"	"	"	"
Y40	"	16,2, 32	11,22,26,31,38	37	"	"	"	"	"	"
Y41	"	24,11, 32	2,26,31,37,38	22	"	"	"	"	"	"
Y42	"	28,11, 32	2,22,31,37,38	26	"	"	"	"	"	"
Y43	"	35,11 32	2,22,26,37,38	31	"	"	"	"	"	"
Y44	"	39,11, 32	2,22,26,31,38	37	"	"	"	"	"	"
Y45	"	5,2, 11,32	26,31,37,38	22	"	"	"	"	"	"
Y46	"	8,2, 11,32	22,31,37,38	26	"	"	"	"	"	"
Y47	"	34,2, 11,32	22,26,37,38	31	"	"	"	"	"	"
Y48	"	19,2, 11,32	22,26,31,38	37	"	"	"	"	"	"
Y49	"	4,32, 38	2,11,26,31,37	22	18	"	"	"	"	"
Y50	"	9,32, 38	2,11,22,31,37	26	"	"	"	"	"	"
Y51	"	15,32, 38	2,11,22,26,37	31	"	"	"	"	"	"

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* TABLE XXV. Propagation delay time (strobe to output, t_{PHZ}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y52	Propagation delay time, strobe to output, (t_{PHZ}), V_{DD} = 5 V	20,32, 38	2,11,22,26,31	37	18	Open	70	700	ns	25,26
Y53	"	23,2, 32,38	11,26,31,37	22	"	"	"	"	"	"
Y54	"	25,2, 32,38	11,22,31,37	26	"	"	"	"	"	"
Y55	"	33,2, 32,38	11,22,26,37	31	"	"	"	"	"	"
Y56	"	16,2, 32,38	11,22,26,31	37	"	"	"	"	"	"
Y57	"	24,11, 32,38	2,26,31,37	22	"	"	"	"	"	"
Y58	"	28,11, 32,38	2,22,31,37	26	"	"	"	"	"	"
Y59	"	35,11, 32,38	2,22,26,37	31	"	"	"	"	"	"
Y60	"	39,11, 32,38	2,22,26,31	37	"	"	"	"	"	"
Y61	"	5,2, 11,32, 38	26,31,37	22	"	"	"	"	"	"
Y62	"	8,2, 11,32, 38	22,31,37	26	"	"	"	"	"	"
Y63	"	34,2, 11,32, 38	22,26,37	31	"	"	"	"	"	"
Y64	"	19,2, 11,32, 38	22,26,31	37	"	"	"	"	"	"

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* TABLE XXVI. Propagation delay time (strobe to output, t_{PZH}) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Z1	Propagation delay time, strobe to output (t_{PZH}), V_{DD} = 5 V	4	2,11,32,38,26,31,37	22	14	Open	60	600	ns	25,27
Z2	"	9	2,11,22,31,32,37,38	26	"	"	"	"	"	"
Z3	"	15	2,11,22,26,32,37,38	31	"	"	"	"	"	"
Z4	"	20	2,11,22,26,31,32,38	37	"	"	"	"	"	"
Z5	"	23,2	11,26,31,32,37,38	22	"	"	"	"	"	"
Z6	"	25,2	11,22,31,32,37,38	26	"	"	"	"	"	"
Z7	"	33,2	11,22,26,32,37,38	31	"	"	"	"	"	"
Z8	"	16,2	11,22,26,31,32,38	37	"	"	"	"	"	"
Z9	"	24,11	2,26,31,32,37,38	22	"	"	"	"	"	"
Z10	"	28,11	2,22,31,32,37,38	26	"	"	"	"	"	"
Z11	"	35,11	2,22,26,32,37,38	31	"	"	"	"	"	"
Z12	"	39,11	2,22,26,31,32,38	37	"	"	"	"	"	"
Z13	"	5,2,11	26,31,32,37,38	22	"	"	"	"	"	"
Z14	"	8,2,11	22,31,32,37,38	26	"	"	"	"	"	"
Z15	"	34,2,11	22,26,32,37,38	31	"	"	"	"	"	"
Z16	"	19,2,11	22,26,31,32,38	37	"	"	"	"	"	"
Z17	"	4,38	2,11,26,31,32,37	22	27	"	"	"	"	"
Z18	"	9,38	2,11,22,31,32,37	26	"	"	"	"	"	"
Z19	"	15,38	2,11,22,26,32,37	31	"	"	"	"	"	"
Z20	"	20,38	2,11,22,26,31,32	37	"	"	"	"	"	"

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* TABLE XXVI. Propagation delay time (strobe to output, t_{PZH}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Z21	Propagation delay time, strobe to output (t_{PZH}), V_{DD} = 5 V	23,2, 38	11,26,31,32,37	22	27	Open	60	600	ns	25,27
Z22	"	25,2, 38	11,22,31,32,37	26	"	"	"	"	"	"
Z23	"	33,2, 38	11,22,26,32,37	31	"	"	"	"	"	"
Z24	"	16,2, 38	11,22,26,31,32	37	"	"	"	"	"	"
Z25	"	24,11 38	2,26,31,32,37	22	"	"	"	"	"	"
Z26	"	28,11 38	2,22,31,32,37	26	"	"	"	"	"	"
Z27	"	35,11, 38	2,22,26,32,37	31	"	"	"	"	"	"
Z28	"	39,11, 38	2,22,26,31,32	37	"	"	"	"	"	"
Z29	"	5,2, 11,38	26,31,32,37	22	"	"	"	"	"	"
Z30	"	8,2, 11,38	22,31,32,37	26	"	"	"	"	"	"
Z31	"	34,2, 11,38	22,26,32,37	31	"	"	"	"	"	"
Z32	"	19,2, 11,38	22,26,31,32	37	"	"	"	"	"	"
Z33	"	4,32	2,11,26,31,37,38	22	29	"	"	"	"	"
Z34	"	9,32	2,11,22,31,37,38	26	"	"	"	"	"	"
Z35	"	15,32	2,11,22,26,37,38	31	"	"	"	"	"	"
Z36	"	20,32	2,11,22,26,31,38	37	"	"	"	"	"	"
Z37	"	23,2, 32	11,26,31,37,38	22	"	"	"	"	"	"

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* TABLE XXVI. Propagation delay time (strobe to output, t_{PZH}) test - Continued.

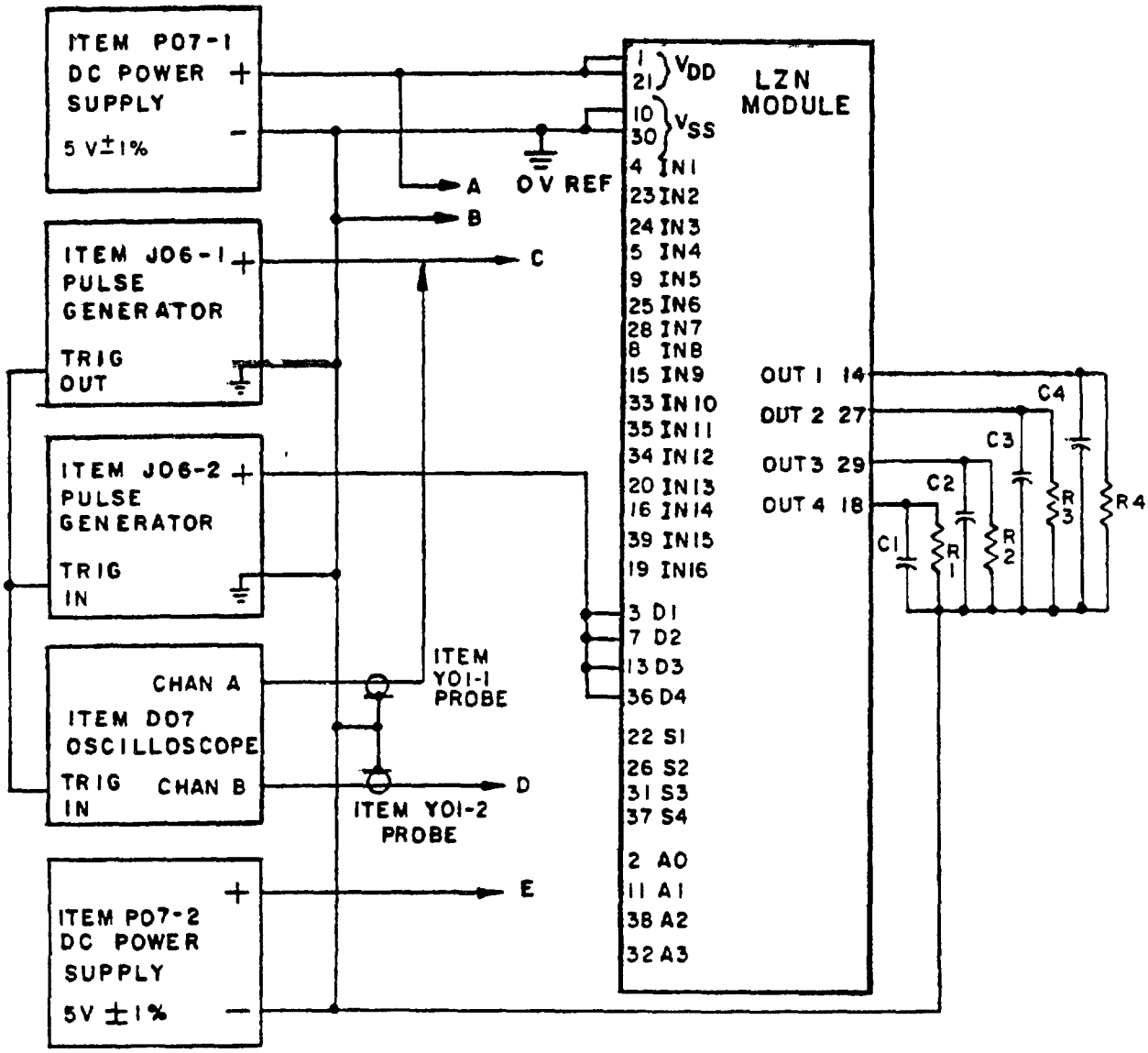
Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Z38	Propagation delay time, strobe to output (t_{PZH}), V_{DD} = 5 V	25,2,32	11,22,31,37,38	26	29	Open	60	600	ns	25,27
Z39	"	33,2,32	11,22,26,37,38	31	"	"	"	"	"	"
Z40	"	16,2,32	11,22,26,31,38	37	"	"	"	"	"	"
Z41	"	24,11,32	2,26,31,37,38	22	"	"	"	"	"	"
Z42	"	28,11,32	2,22,31,37,38	26	"	"	"	"	"	"
Z43	"	35,11 32	2,22,26,37,38	31	"	"	"	"	"	"
Z44	"	39,11,32	2,22,26,31,38	37	"	"	"	"	"	"
Z45	"	5,2,11,32	26,31,37,38	22	"	"	"	"	"	"
Z46	"	8,2,11,32	22,31,37,38	26	"	"	"	"	"	"
Z47	"	34,2,11,32	22,26,37,38	31	"	"	"	"	"	"
Z48	"	19,2,11,32	22,26,31,38	37	"	"	"	"	"	"
Z49	"	4,32,38	2,11,26,31,37	22	18	"	"	"	"	"
Z50	"	9,32,38	2,11,22,31,37	26	"	"	"	"	"	"
Z51	"	15,32,38	2,11,22,26,37	31	"	"	"	"	"	"

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* TABLE XXVI. Propagation delay time (strobe to output, t_{PZH}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Z52	Propagation delay time, strobe to output, (t_{PZH}), V_{DD} = 5 V	20,32,38	2,11,22,26,31	37	18	Open	60	600	ns	25,27
Z53	"	23,2,32,38	11,26,31,37	22	"	"	"	"	"	"
Z54	"	25,2,32,38	11,22,31,37	26	"	"	"	"	"	"
Z55	"	33,2,32,38	11,22,26,37	31	"	"	"	"	"	"
Z56	"	16,2,32,38	11,22,26,31	37	"	"	"	"	"	"
Z57	"	24,11,32,38	2,26,31,37	22	"	"	"	"	"	"
Z58	"	28,11,32,38	2,22,31,37	26	"	"	"	"	"	"
Z59	"	35,11,32,38	2,22,26,37	31	"	"	"	"	"	"
Z60	"	39,11,32,38	2,22,26,31	37	"	"	"	"	"	"
Z61	"	5,2,11,32,38	26,31,37	22	"	"	"	"	"	"
Z62	"	8,2,11,32,38	22,31,37	26	"	"	"	"	"	"
Z63	"	34,2,11,32,38	22,26,37	31	"	"	"	"	"	"
Z64	"	19,2,11,32,38	22,26,31	37	"	"	"	"	"	"

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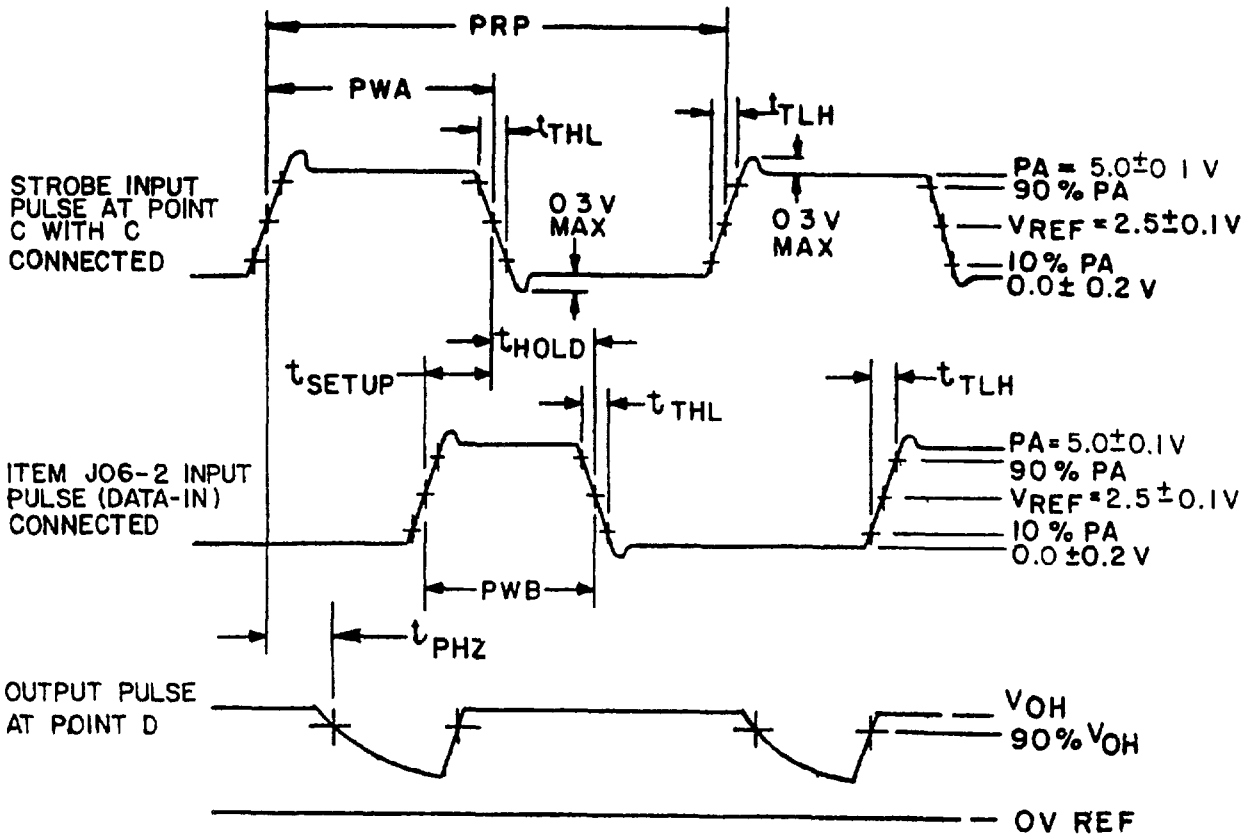
CAUTION:

Observe the procedures of 6.6.6.

NOTES:
1. R1-R4 = 1 kΩ ±1%, 1/8 W.
2. C1-C4 = 47 pF ±5% (including probe and parasistic capacitance).
3. For accurate test results each analog switch must initially be OFF. Set the switches to OFF prior to each test condition by performing the procedure of figure 6 (set V_{DD} = 5 V ±1%, logic high level = same as V_{DD}, and logic low level = 0 V).

* FIGURE 25. Propagation delay time (strobe to otuput, t_{PHZ} and t_{PZH}) test circuit for tables XXV and XXVI.

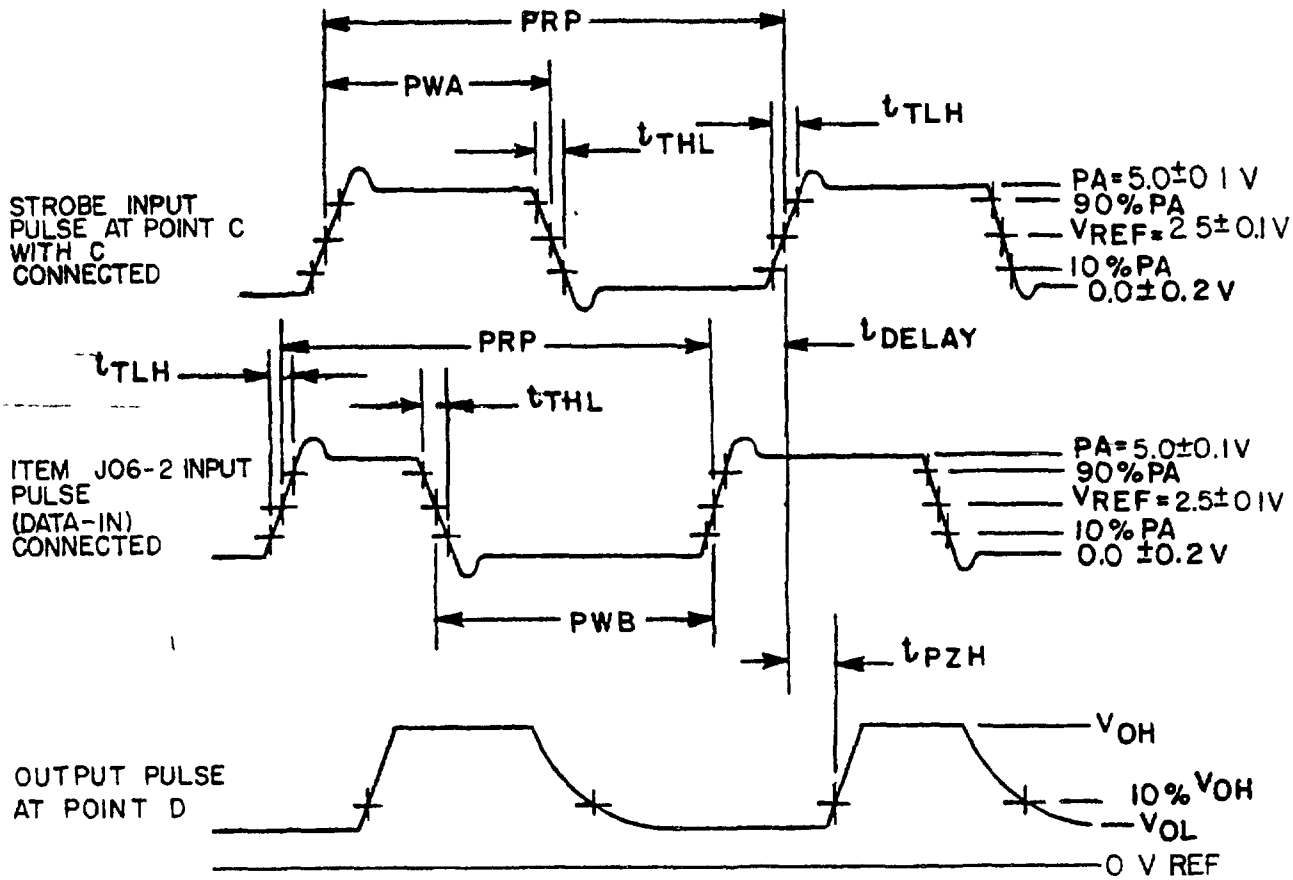
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NOTES:
1. Input pulse parameters:
   PRP = 2.0 ±0.1 µs.
   PWA = 1.00 ±0.05 µs.
   PWB = 615 ±21 ns.
   $t_{TLH} = t_{THL}$ = 20 ±4 ns.
   t_{SETUP} = 200 ±6 ns.
   t_{HOLD} = 415 ±15 ns.
2. Before making any measurements, the setup and hold times must be verified.
3. The lead length of electrical conductors between resistors and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to 0 V reference.

* FIGURE 26. Propagation delay time (strobe to output, t_{PHZ}) test circuit waveforms for table XXV.

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NOTES:
1. Input pulse parameters.
   PRP = 2.4 ±0.1 µs.
   PWA = 1.40 ±0.05 µs.
   PWB = 1.40 ±0.05 µs.
   $t_{TLH} = t_{THL}$ = 20 ±4 ns.
   t_{DELAY} = 220 ±20 ns.
2. Before making any measurements t_{DELAY} must be verified.
3. The lead length of electrical conductors between resistors and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to 0 V reference except V_{OH} which is referenced to V_{OL}.
5. Before making the measurements, verify that the output pulse high level is ≥ 3.5 V and the low level is ≤ 1 V.

* FIGURE 27. Propagation delay time (strobe to output, t_{PZH}) test circuit waveforms for table XXVI.

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* TABLE XXVII. Propagation delay time (data-in to output, t_{PHZ}) test. 1/

Test condition	Parameter	Connect					Initial limits 25°c ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
AA1	Propagation delay time, data-in to output (t_{PHZ}), V_{DD} = 5 V	24,11	7,13,36,2,38,32	3	14	Open	60	600	ns	28,29
AA2	"	28,11,38	3,13,36,2,32	7	27	"	"	"	"	"
AA3	"	35,11,32	3,7,36,2,38	13	29	"	"	"	"	"
AA4	"	39,11,38,32	3,7,13,2	36	18	"	"	"	"	"

1/ This test may be performed concurrently with test conditions BB1-BB4.

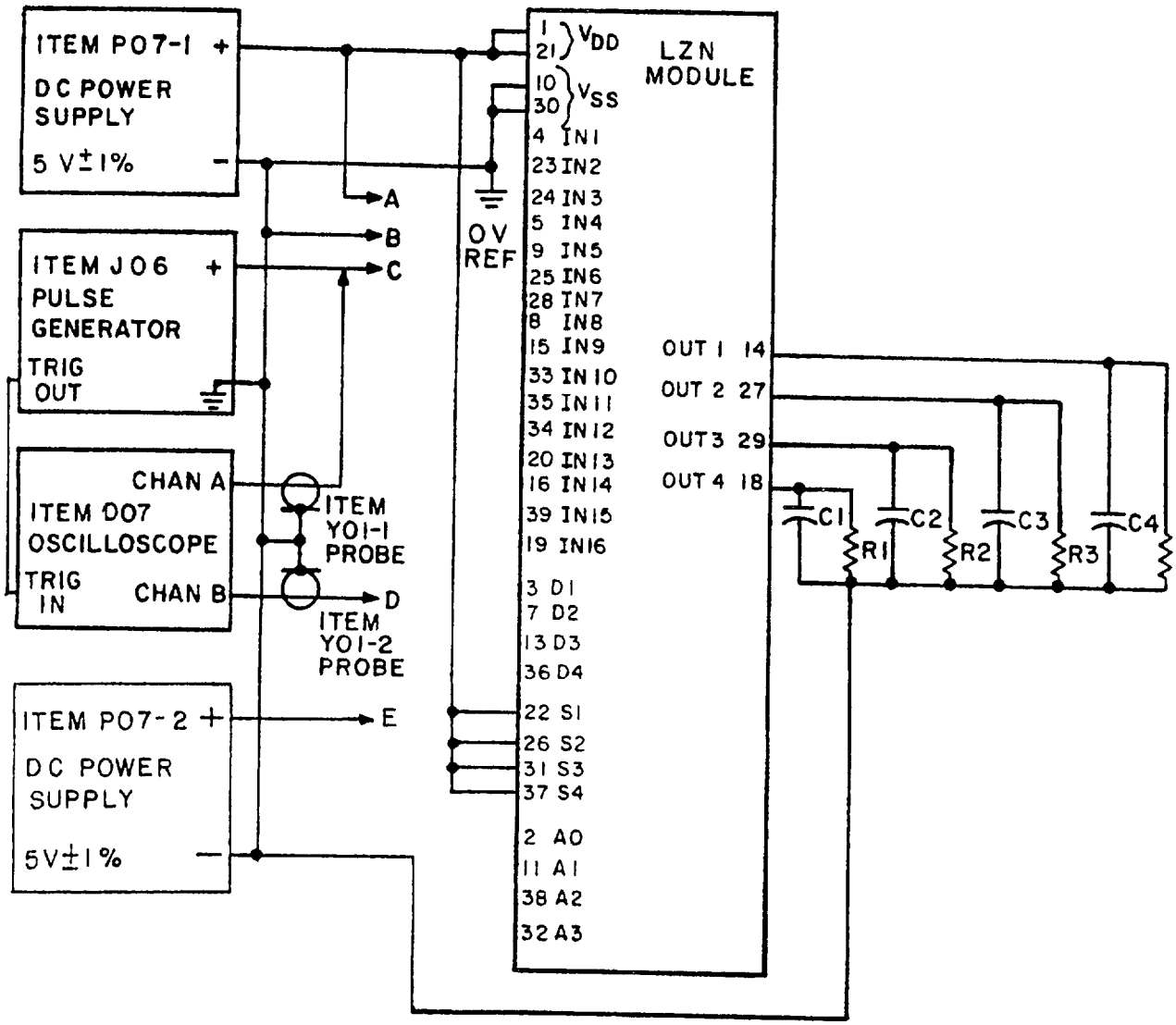
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* TABLE XXVIII. Propagation delay time (data-in to output, t_{PZH}) test. 1/

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
BB1	Propagation delay time, data-in to output (t_{PZH}), V_{DD} = 5 V	24,11	7,13,36,2,38,32	3	14	Open	60	600	ns	28,29
BB2	"	28,11,38	3,13,36,2,32	7	27	"	"	"	"	"
BB3	"	35,11,32	3,7,36,2,38	13	29	"	"	"	"	"
BB4	"	39,11,38,32	3,7,13,2	36	18	"	"	"	"	"

1/ This test may be performed concurrently with test conditions AA1-AA4.

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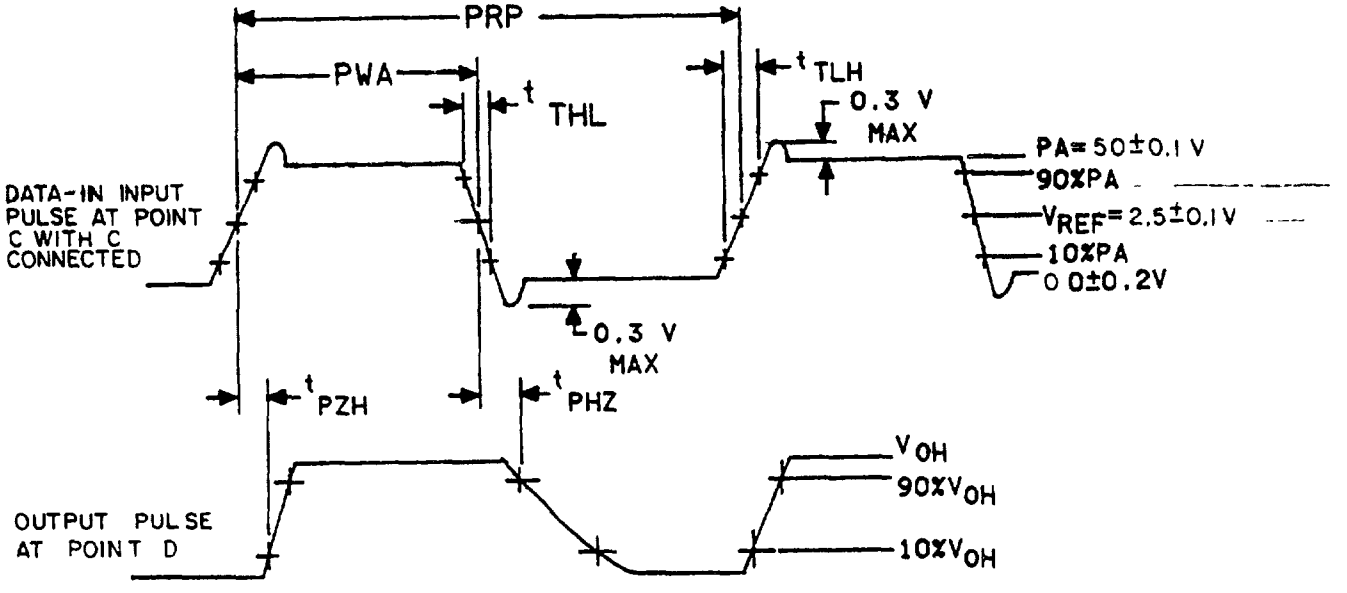
CAUTION:

Observe the procedures of 6.6.6.

NOTES:
1. R1-R4 = 1 kΩ ±1%, 1/8 W.
2. C1-C4 = 47 pF ±5% (including probe and parasitic capacitance).
3. For accurate test results each analog switch must initially be OFF. Set the switches to OFF prior to each test condition by performing the procedure of figure 6 (set V_{DD} = 5 V ±1%, logic high level = same as V_{DD} and logic low level = 0 V).

* FIGURE 28. Propagation delay time (data-in to output, t_{PHZ} and t_{PZH}) test circuit for tables XXVII and XXVIII.

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NOTES:
1. Input pulse parameters:
   PRP = 2.0 ±0.1 µs.
   PWA = 1.00 ±0.05 µs.
   $t_{TLH} = t_{THL}$ = 20 ±4 ns.
2. The lead length of electrical conductors between resistors and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
3. All voltages are referenced to 0 V reference.

* FIGURE 29. Propagation delay time (data-in to output, t_{PHZ} and t_{PZH}) test circuit waveforms for tables XXVII and XXVIII.

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